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Technical Report No. 68 (Revised 2024)

Risk-Based Approach for Prevention and Management of Drug Shortages

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Risk-Based Approach for Prevention and Management of Drug Shortages

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Technical Report No. 68 (Revised 2024)

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1.0 Introduction

PDA *Technical Report No. 68: Risk-Based Approach for Prevention and Management of Drug Shortages* was originally developed in 2014 as part of the interassociation collaborative response to the European Medicines Agency's (EMA) initiative on medicinal product shortages caused by manufacturing, quality, and/or good manufacturing practice (GMP) issues. This revision updates the 2014 version to recognize emerging regulatory expectations and current terminology. The risk-based approach and concepts in the original technical report are broadly recognized and remain unchanged.

This revision recognizes the advances made in pharmaceutical manufacturing and aligns the text of TR-68 with the current regulatory environment, for example, the International Council for Harmonisation (ICH) *Quality Guideline Q9(R1): Quality Risk Management*, ICH *Quality Guideline Q12: Technical and Regulatory Considerations for Pharmaceutical Product Lifecycle Management,* regional regulations and guidelines, and terminologies used for drug shortages that have since been published. Examples for preventive measures to avoid drug shortages consider lessons learned since TR-68 was first published in 2014. Elements in the templates and forms included in this revision have been updated to reflect current regulatory expectations and terminology used in the European Union and its Member States and in the United States, while it also supports compliance with global expectations on managing and preventing drug shortages.

Sustainable access to safe, efficacious, and high-quality products every time a patient needs them is just as important as the therapeutic advances that have transformed the lives and survival of patients worldwide. The ICH Q9(R1) defines harm as "damage to health, including damage that can occur from loss of product quality or availability" ***(1)***. Ensuring uninterrupted availability of and access to products, especially medically necessary, lifesaving or life sustaining critical medicines that do not have effective alternatives, is essential.

1.1 Background

Drug shortages have become more frequent, more severe, and longer-lasting globally since the early 2000s for a variety of reasons**.** As globalization increases, the complexity of supply chains grows, global and regional regulatory expectations evolve, and economic and business motivators change. The drivers behind drug shortages in different geographic areas and countries vary, and those causes are usually multifactorial. The range of causes includes financial, economic, and business concerns, increased or unanticipated market demand (e.g., pandemic), evolving regulatory expectations, manufacturing and quality issues, capacity constraints, lack of redundancy, and supply chain issues ***(2-5)***. Recognition has evolved that the multifactorial root causes for shortages extend beyond manufacturing and quality issues, as has the need for multistakeholder interaction and collaboration to address the local issue of a shortage in a multiregional environment.

The emerging global challenge of supply disruptions is drawing greater focus from the pharmaceutical industry, regulators, legislators, healthcare providers, patient organizations, and patients, and drug shortages are now covered in the media to a much larger extent. ICH Q9(R1) places greater emphasis on product availability risks—a noteworthy change from the original guideline—in the context of quality and manufacturing issues, including GMP noncompliance, that can be a "significant cause of product availability issues (e.g., product shortages)." Furthermore, while "manufacturing and supply chain diversity can be enablers of product availability, increasingly complex supply chains lead to interdependencies that can introduce systemic quality/manufacturing risks impacting supply chain robustness" ***(1)***. Thus, the interests of patients are served by risk-based drug shortage prevention and mitigation activities that help to proactively manage supply chain complexities and ensure timely availability of needed medicinal products.

The legislative and regulatory processes for shortage-related issues continue to evolve. In 2012, EMA published a reflection paper and implementation plan for drug shortages that affect patients. EMA followed

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up by holding workshops with stakeholders in October 2013 and again in October 2015 ***(6-8)***. In 2016, the European Heads of Medicines Agencies (HMA) and EMA established a joint task force to develop and coordinate actions for better prevention, identification, management, and communication of issues affecting the availability of medicines to improve the continuity of medicines supply across Europe. In 2020, the European Commission's Pharmaceutical Strategy further elaborated upon the need for proactive measures to address the issue of drug shortages ***(9)***, and in March 2023, the HMA/EMA Task Force held a third workshop to inform and update stakeholders on the HMA/EMA Task Force's deliverables and activities. This workshop provided an opportunity for the diverse groups to share their perspectives and discuss their respective ongoing or planned initiatives to address medicines availability issues ***(10)***.

The focus on drug shortages has also increased in the United States. In 2013, the U.S. Food and Drug Administration (FDA) established a Drug Shortage Task Force, which published a strategic plan for preventing and reducing drug shortages. The task force continues to identify the root causes of drug shortages and advance long-term solutions, reporting annually to Congress. As in many countries, notification of a potential drug shortage is a legal requirement in the United States, and early and timely notifications has reduced the number of shortages by enabling manufacturers and health authorities to jointly take steps to minimize supply disruption ***(11)***. The annual reports to Congress for calendar years 2021 and 2022 included the impact of the increased risk of shortages during the COVID-19 pandemic and the actions taken to address them; both reports highlighted the crucial role early notifications played to reduce the impact and threat of shortages ***(12, 13)***. In 2021, the White House published a report, *Building Resilient Supply Chains, Revitalizing American Manufacturing, And Fostering Broad-Based Growth,* that emphasized the importance of increasing the resilience and transparency of global supply chains, incentivizing quality management maturity, and increasing economic sustainability of U.S. manufacturing ***(14)***. A 2023 Majority Staff Report of the U.S. Senate Committee on Homeland Security and Governmental Affairs, *Short Supply: The Health and National Security Risks of Drug Shortages*, concluded that active drug shortages had risen by 30% between 2021 and 2022, lasted one-and-a-half years longer on average, and impacted patient care. According to the report, overreliance on foreign resources was deemed a national security risk, FDA lacked critical information to mitigate shortages, and both industry and regulators lacked end-to-end supply chain visibility ***(5)***.

Consequently, health authorities and the pharmaceutical industry have undertaken more proactive efforts and increased multistakeholder engagement to recognize the importance of managing stakeholder interdependencies. There is greater awareness of how the unintended consequences of actions taken by any one stakeholder group can negatively impact other stakeholder processes. The European Union, United States, and other countries acknowledge the broad range of multifactorial root causes, recognize the need for multistakeholder interactions and collaborative solutions, appreciate the importance of end-to-end supply chain visibility, and share a common understanding and political will to significantly reduce shortages. The awareness and acknowledgment of this complex ecosystem is gradually growing, as depicted in **Figure 1.0-1**.

It is essential for the senior management of a firm that is the Marketing Authorization Holder (MAH) to have control of and be accountable for ensuring the uninterrupted supply of their products as long as it is within their control ***(15)***. Senior Management decisions impact not only a firm's ability to cause shortages but also to prevent and recover from shortages in an expedient manner. As stated in ICH Q9 (R1), the principles of quality risk management (QRM) that enable risk-based decision-making by both companies and health authorities are fundamental for the proactive prevention of drug shortages.

This concept of prevention and the use of risk-based principles cannot be overstated in terms of importance. The November 2020 ICH Concept Paper that formed the basis for the revision of ICH Q9 stated that "addressing lifecycle risks to manufacturing reliability and quality assurance is the foundation for supply predictability" ***(16)***.

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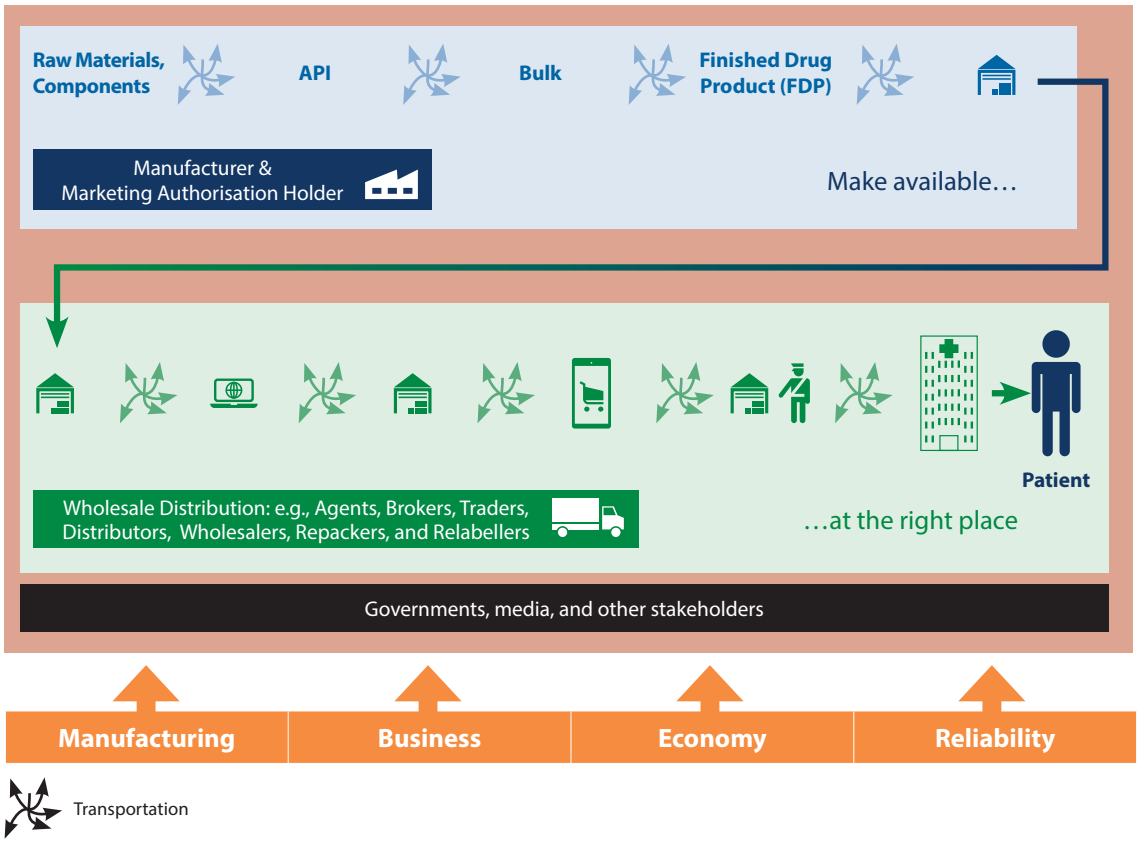

Figure 1.0-1 Mock-up of the Holistic Process for Manufacturing and Supply, and the Types of Influencing Factors

1.2 Purpose and Scope

This revision of TR-68 addresses the following aspects for proactive prevention of drug shortages and provides useful tools to assist in preparation:

- A holistic, risk-based framework at a product level for the prevention and management of drug shortages caused by manufacturing and quality issues
- A risk triage model that can be used to assess drug shortage risks and implement appropriate controls in the end-to-end value chain for the manufacturing and distribution of a product (**Section 11.0**, **Appendix 1**)
- Templates for developing a Drug Shortage Risk Register and a Drug Shortage Prevention and Response Plan at a product level (**Section 12.0**, **Appendix 2**)

The concepts and approach presented in this report may be adopted and applied by other stakeholders in the entire supply chain, such as wholesale distributors, as well as suppliers and vendors of materials used in processing, for example, filters, single-use bags, flow paths, resins, and solvents.

TR-68 focuses on addressing shortages caused by manufacturing and quality issues toward batch release. The next step in the supply chain after manufacturing and release is usually called the "handover, to the first economic customer," after which the MAH often has limited-to-no visibility (e.g., shortages related to issues that arise in distribution channels, pharmacies, or hospitals). Shortages after the "handover, to the first economic customer" are out of scope of this technical report. Nevertheless, there may be situations where the MAH is responsible for distribution as well and, therefore, is responsible for mitigating shortages in these instances. TR-68 does apply in these situations.

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The framework provided in TR-68 also supports and enables MAHs and manufacturers to meet the requirements of Section 506C of the *U.S. Federal Food, Drug, and Cosmetic Act* and Article 13 of EU *Commission Directive 2003/94/EC Laying Down the Principles and Guidelines of Good Manufacturing Practice in Respect of Medicinal Products for Human Use and Investigational Medicinal Products for Human Use* to notify relevant health authorities in the event of a drug shortage ***(17, 18)***.

Development of PDA's risk-based approach for prevention and management of drug shortages at a product level began in 2012 as part of the Paradigm Change in Manufacturing Operations (PCMO®) project and incorporated input from the EMA-sanctioned industry interassociation task force and multistakeholder events. An overview of the risk-based approach was published in April 2014 in a *PDA Letter* article and is elaborated in detail in this technical report ***(19)***. PDA also held a drug shortage workshop in September 2014 and participated in various HMA/EMA multistakeholder workshops, including the one held in 2023; key outcomes from the workshop discussions have been incorporated into this revision of TR-68. Additionally, PDA's *Technical Report No. 54: Implementation of Quality Risk Management for Pharmaceutical and Biotechnology Manufacturing Operations* and its series of best-practice and companion case-study documents provide detailed guidance and practical application on how to implement QRM for the management of product quality risks in manufacturing operations, supply chains, and operations of third parties (e.g., suppliers, contract manufacturers). *PDA Technical Report No. 59: Utilization of Statistical Methods for Production Monitoring* describes additional risk-based activities that use statistical methodologies for quality control and manufacturing.

2.0 Glossary and Abbreviations

Critical Medicine

Critical medicines, sometimes called "essential medicines" or "medically necessary drugs," are considered critical for public health, taking into consideration the therapeutic indication of the medicines, the availability of alternative treatments, and the vulnerability of the supply chain ***(20)***.

Corrective Action and Preventative Action (CAPA)

Action to eliminate the cause of a detected nonconformity or other undesirable situation.

Critical Shortage

A shortage where no alternative treatment is available ***(10)***.

Drug Shortage or Shortage

A shortage of a drug or medicinal product that occurs when supply does not meet demand at a national level, where a:

> **Shortage** allows for identification of current, impeding, or anticipated disruption of supply of a medicinal product.

Supply refers to the total volume of stock of the individual medicinal product that is placed on the market by the MAH.

Demand relates to the request for a medicinal product by a healthcare professional, veterinarian, or patient in response to a clinical need. For demand to be satisfactorily met, the medicinal product will need to be acquired in time and in sufficient quantity to allow continuity of best care of patients or animals.

National level refers to the situation in a specific country, that is, if there is an insufficient supply of medicine to meet the demands of the country overall. Logistics-related issues leading to regional supply disruption of a medicinal product (e.g., delivery difficulties, national redistribution of stock) are a short-term and localized problem and should not be taken into account ***(21)***.

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Drug Shortage Prevention & Response Plan
A document that provides a structured action plan to proactively prevent drug shortages and also respond to a shortage in the event that one occurs. This also may also be known as a "Drug Shortage Prevention and Mitigation Plan."

Drug Shortage Risk Register
A single source of information on risks that can result in drug shortages, associated risk levels, risk control actions with owners, status, due dates, and residual risk after appropriate risk control actions have been taken.

Harm
Damage to health, including the damage that can occur from loss of product quality or availability ***(1)***.

Knowledge Management
Systematic approach to acquiring, analyzing, storing, and disseminating information related to products, manufacturing processes, and components ***(22)***.

Life supporting or Life sustaining Drug
Life supporting or life sustaining drug is a term used to describe a product that is essential to, or that yields information that is essential to, the restoration or continuation of a bodily function that is important to the continuation of human life ***(23)***.

Meaningful Disruption
A meaningful disruption is a change in production that is reasonably likely to lead to a reduction in the supply of a drug by a manufacturer that is more than negligible and affects the ability of the manufacturer to fill orders or meet expected demand for its product. A meaningful disruption is not an interruption in manufacturing due to matters such as routine maintenance and does not include insignificant changes in manufacturing so long as the manufacturer expects to resume operations in a short period of time ***(23)***.

Medically Necessary Drug
Any drug product used to treat or prevent a serious disease or medical condition for which there is no other adequately available drug product that is judged by medical staff to be an appropriate substitute ***(20)***.

Potential Drug Shortage
A potential drug shortage is described as the occurrence of internal or external situations (single or in a combination of both), which could result in an interruption of supplies of a medicinal product, if not properly addressed and controlled ***(24)*.**

Quality Risk Management
A systematic process for the assessment, control, communication, and review of risks to the quality of the drug (medicinal) product across the product lifecycle ***(1)***.

Residual Risk
Risk remaining after risk control measures have been implemented ***(1)***. In the context of drug shortages, this is considered to be the risk of a shortage after preventive and other risk control measures have been implemented.

Risk
The combination of the probability of occurrence of harm and the severity of that harm, where harm is defined as "damage to health, including the damage that can occur from loss of product quality or availability" ***(1)***.

Risk-based Triage for Drug Shortages
A process of assessing and assigning priorities for managing drug shortage risks based on criticality and impact to patients.

Risk Priority Level
A relative priority ranking assigned to a risk based on a combination of (a) therapeutic use of a product and patient impact due to product unavailability, (b) availability of alternatives, and (c) likelihood of a shortage.

Value Stream Map
A tool used to document, analyze, understand, and improve the flow of information or materials required to produce a product or service for a customer as it makes its way through the value stream (a set of actions that take place to add value for customers from the initial request through realization of value by the customer).

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2.1 Abbreviations

ANDA Abbreviated New Drug Application

API Active Pharmaceutical Ingredient

CAPA Corrective Action and Preventative Action

EMA European Medicines Agency

FDA U.S. Food and Drug Administration

GDP Good Distribution Practice

GMP Good Manufacturing Practice

ICH International Council for Harmonisation

MAH Marketing Authorization Holder

NDA New Drug Application

PAC Post-Approval Change

PQS Pharmaceutical Quality System

QRM Quality Risk Management

VSM Value Stream Mapping

WHO World Health Organization

3.0 Recognizing Drug Shortages as a Global Problem

The primary objective in the management of drug shortages is preventing harm to patients. In any situation, it is important to maintain the continuity of supply such that all patients receive the medicines they need every time they need them to manage their medical condition. The risk of potential drug shortages should be considered when making decisions related to the supply and/or manufacture of drug products and their corresponding active substances. This technical report addresses how drug shortages caused by manufacturing and quality issues can be anticipated and avoided using a risk-based approach at a product level.

A good example of a shortage that had global impact and needed to be managed by multiple health authorities was "The Great Acetonitrile Shortage" in 2008-2009. Acetonitrile is used as a solvent in many analytical procedures that are included in submissions for the approval of drug products and in chemical reactions for the manufacture of active pharmaceutical ingredients (API). The two main manufacturers of this solvent, both located in the same area in China, were required to stop manufacturing acetonitrile to reduce air pollution for the 2008 Summer Olympics. A few months later, in September of 2008, acetonitrile production in Texas was interrupted due to hurricanes in the Gulf of Mexico. The combination of these two events caused shortages of several APIs and drug products. Only a few companies were aware that the product would not be manufactured for a year and, subsequently, had built up their inventories of the product in advance. Most regulatory agencies accepted alternative analytical methods on a short-term basis. The FDA also allowed changes to existing validated test methods within an approved application, either a new drug application (NDA) or an abbreviated new drug application (ANDA), to accommodate using less acetonitrile or an alternative solvent to acetonitrile. The Agency allowed companies to process this as a minor change in the Annual Report for the given drug application as long as the change met the criteria stated in the FDA *Guidance for Industry: Changes to an Approved NDA or ANDA* and *Code of Federal Regulations* [(21CFR314.70(d)(2)(vii)] ***(25, 26)***.

This example illustrates that, in some cases, drug shortages can occur very suddenly and for reasons that are difficult to predict. Common antibiotics such as amoxicillin have also experienced shortages

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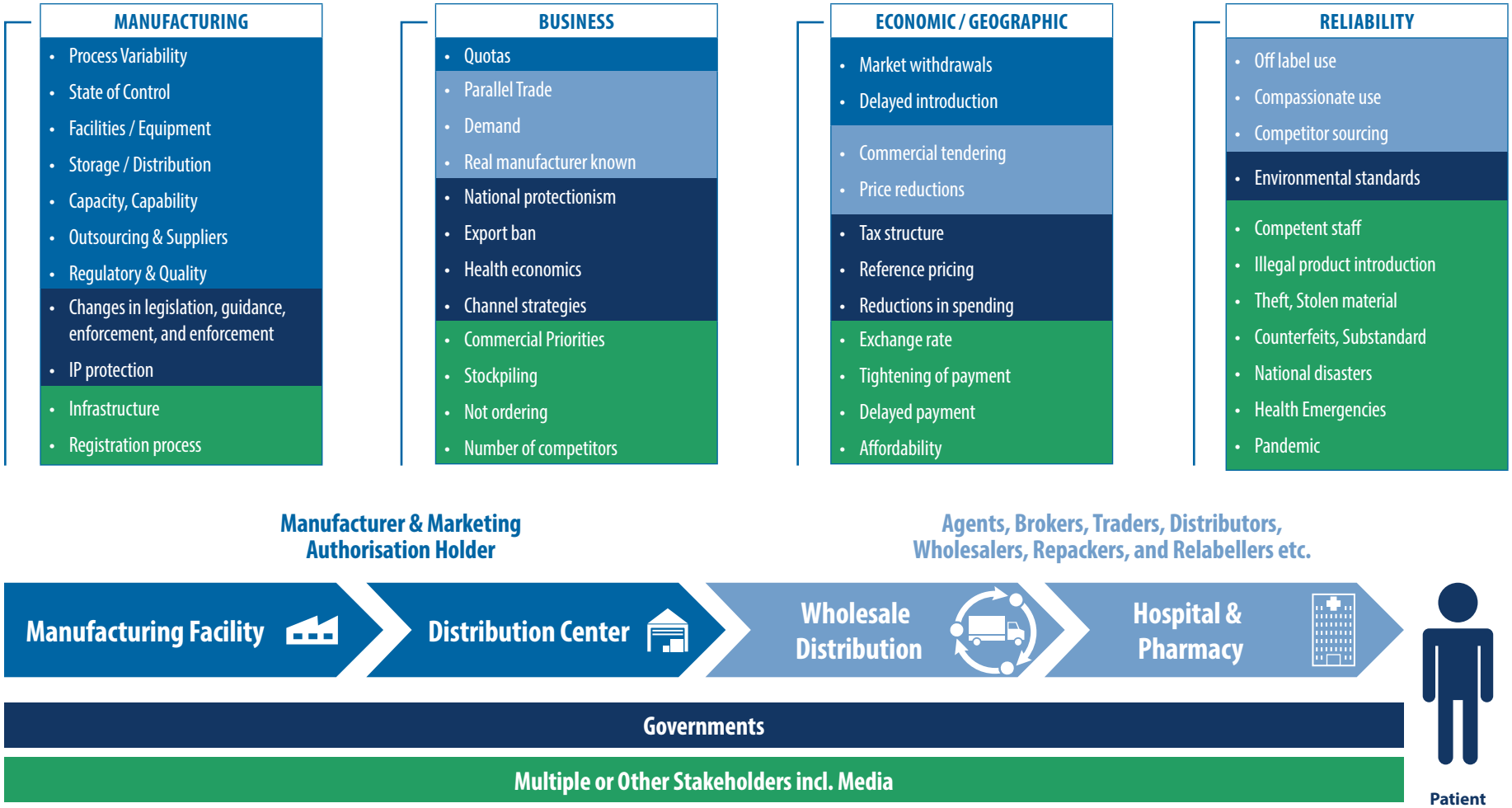

Figure 3.0-1 Examples of the Diversity of Factors and Stakeholders that Could Contribute to Drug Shortages

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globally due to supply-chain disruptions. Hence, it is important for companies and regulators alike to do their best to anticipate potential drug shortages and prevent them from happening. Improved surveillance and communications through forums and consortiums, such as the International Supply Chain Consortium (Rx-360), can help in managing widespread impact in such instances ***(27)***.

The 2013 Birgli® report identified three principal types of causes for drug shortages—economic, business, and manufacturing supply chain issues—that are still highly relevant today ***(2)***. Since then, an awareness of the diversity of possible root causes for shortages has continued to evolve ***(4)***. **Figure 3.0-1** illustrates the impact across manufacturers, MAHs, wholesale distributors, hospitals and pharmacies, governments, and other stakeholders based on the current understanding of multifactorial root causes for driving drug shortages ***(27)***. It is important to acknowledge the role of the media in public perception of shortages.

Figure 3.0-1 illustrates why it is important to recognize the landscape of stakeholders and stakeholder interdependencies as a complex ecosystem. It underscores the need for appropriate and timely communications among all the players to better manage unintended consequences of actions that any one stakeholder might take to prevent or manage drug shortages.

3.1 Evolution of Terminologies

Though single, internationally harmonized definitions do not currently exist for "drug shortage," "meaningful disruption," and "medically necessary product," various definitions of these terms have been published. The FDA defines drug shortage as "a period of time when the demand or projected demand for the drug in the United States exceeds the drug's supply " ***(28)***. In general, the FDA Drug Shortage Staff focuses on shortages of medically necessary products that have a significant effect on public health ***(18, 23, 29)***. The EMA earlier described drug shortages as "disruptions to the manufacturing supply chain" ***(6, 30)*** and updated the definition of shortages in 2019 to be more specific in terms of the gap between supply and demand ***(21)***. In addition, individual EU member states have local definitions of drug shortages ***(31)***.

The EU's current definition of "critical medicines" also includes the criteria of therapeutic use, availability of alternative(s), and supply-chain vulnerabilities. The World Health Organization (WHO) refers to a "shortage" as a situation where the supply of medicines, health products or vaccines identified as essential by the health system is considered to be insufficient to meet public health and patient needs. WHO indicates that:

> On the demand side: a "shortage" will occur when demand exceeds supply at any point in the supply chain and may ultimately create a "stockout" at the point of appropriate service delivery to the patient if the cause of the shortage cannot be resolved in a timely manner relative to the clinical needs of the patient ***(32)***.

The understanding of these terms may also vary in some respects by the different stakeholders in the supply chain (MAHs, regulatory authorities, distributors, wholesalers, providers, and patients). However, there is alignment on the concept that these definitions are based on the fundamental importance of ensuring an uninterrupted supply of medicines to prevent harm to patients. They are also aligned with public-service obligations held by MAHs and wholesale distributors in relation to ensuring the supply of medicinal products in the markets in which they operate.

3.2 Reporting Requirements and Current Regulatory Landscapes Related to Drug Shortages

Reporting potential shortages to relevant health authorities is a legal requirement. In the United States, Section 506C of the *Federal Food, Drug, and Cosmetic Act* establishes notification requirements to

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manufacturers of product discontinuance ***(18)***. Title X of the *Food and Drug Administration Safety and Innovation Act*, enacted in July 2012 to address drug shortages, also led to Section 506C being amended to require annual reporting of drug shortages to the U.S. Congress ***(33)***. However, it has been noted that manufacturers are not required by law to report increased demand or export restrictions for life supporting or life sustaining products ***(5)***.

In the European Union, Article 13 of EU GMP Directive 2017/1572 defines the legal requirement for a manufacturer "to inform the competent authority and, if applicable, the MAH of any defect that could result in a recall or an abnormal restriction on supply" ***(34)***. Local legislation implementing this directive in the member states has demonstrated individual approaches being followed by each country. Additionally, in August 2014, *Chapter 5: Production* of *EudraLex – Volume 4 – Guidelines for Good Manufacturing Practices for Medicinal Products for Human and Veterinary Use, Part 1,* was updated to introduce guidance in relation to supply restrictions and their notification to the MAH (Sec. 5.71). It specified that such notifications "should be done in a timely manner to facilitate reporting of the restriction in supply by the MAH, to the relevant competent authorities" ***(35)***. Other health authorities (e.g., individual EU member states, Health Canada, Swissmedic, Therapeutic Goods Administration) are also implementing regulatory requirements in this regard. ICH Q9(R1) further highlights the need for appropriate and timely management of health authority notification.

While reporting requirements and mechanisms for drug shortages vary across different health authorities and are subject to varying legal requirements, the expectation of visibility to shortage information has increased over the last several years. More information can be found at the websites of the individual health authorities on local drug shortage situations.[1] Other associations, such as the American Society of Health System Pharmacists, also may provide information on drug shortages and their management through their websites. Health Canada has provided a multistakeholder toolkit that provides strategies and tools for key players in the supply chain. This list demonstrates the variety of stakeholders (e.g., manufacturers, group purchasing organizations, regional health authorities, pharmacists, health care professionals, federal government among others) that should be responsible for addressing drug shortages at specific stages in the supply chain ***(36, 37)***. The differences in reporting requirements and approaches in different countries continue to make it challenging to draw meaningful comparisons of data on drug shortages or to quantify the problem at a global or regional level. The European Federation of Pharmaceutical Industries and Associations (EFPIA), European Generic and Biosimilar Medicines Association (EGA), Association of the European Self-Care Industry (AESGP), and Plasma Protein Therapeutics Association (PPTA) jointly developed a harmonized communication framework for potential drug shortages, which includes harmonized trigger points for notifying competent authorities of drug shortages that are due to quality and manufacturing problems. The paper also provides a detailed listing of information systems from EMA and some EU member states that provide information on drug shortages ***(38)***. In 2019, EMA also published a good practice guidance for EU national competent authorities to communicate medicines availability issues to the public ***(39)***.

In the United States, Congress legislates the FDA's authority to act in response to drug shortages. In October 2013, FDA published its *Strategic Plan for Preventing and Reducing Drug Shortages*. It addresses mitigation strategies for manufacturers, such as notifying the FDA of potential shortages promptly so that manufacturers and the FDA can jointly take steps to reduce supply disruption.

Another element of the FDA's drug shortage strategic plan focuses on preventive strategies, such as providing positive incentives for manufacturing and quality improvements and using proactive risk-based approaches for early identification and prevention of manufacturing and quality problems ***(11)***.

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[1] Refer to individual country websites as web addresses change frequently.

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The annual *FDA Report to Congress* on drug shortages summarizes discussions and actions related to drug shortages in the United States and Puerto Rico, providing appropriate guidance on how to manage drug shortages to minimize their effects on patients ***(12, 13, 23)***.

In 2012, the EMA published a *Reflection Paper on Medicinal Product Supply Shortages Caused by Manufacturing/Good Manufacturing Practice Compliance Problems* and a supporting *Implementation Plan 2012-2015*. In 2019, HMA and EMA published a *Guidance on Detection and Notification of Shortages of Medicinal Products for Marketing Authorization Holders (MAHs) in the Union (EEA)*, which also recommends that MAHs utilize PDA TR-68 as a reference.

The increased transparency of shortages, coupled with the mandatory requirements for early notification to health authorities of potential shortages, has allowed authorities and MAHs to work together more proactively to control and prevent potential drug shortage risks before they result in a supply disruption**.**

Although not all shortages result in patient harm, health authorities try to determine the causes of shortages that do affect patients and "near misses" that could have resulted in patient harm. The objective of doing so is to be prepared to proactively implement actions that can prevent recurrences. For this reason, FDA, EMA, and many other health authorities have taken the crisis around drug shortages seriously and regulations continue to evolve to heighten the attention, sensitivity, and response to shortages.

3.3 Framework for Managing Drug Shortages due to Manufacturing and Quality Problems

As illustrated in **Figure 3.0-1**, the potential causes of drug shortages are often multifactorial and complex. Business, environmental, and economic policies are also important in controlling the risks to drug availability across the end-to-end product value chain, manufacturing being a key step. Therefore, efforts must be concentrated on maintaining uninterrupted manufacturing and distribution and minimizing the risks that packaged drugs will be unavailable.

In the context of product value or supply chain, a drug shortage starts whenever an API, excipient, API starting material, reagent, solvent, component, or even packaging material cannot be made available for the manufacturing process. This can occur due to several reasons, including manufacturing, quality, and GMP compliance issues, inadequate manufacturing capacity to meet increased product demand, lack of redundancy, complexity of the supply chain, evolving global regulatory requirements and the complexity associated with implementing them, and dynamic market conditions. Country- or region-specific import testing requirements can add complexity to supply chains and restrict companies from recovering quickly, potentially turning "near misses" into actual drug shortages.

Manufacturing and quality issues are often listed as a cause of drug shortages in many regions. In this context, shortages might occur as a result of issues related to such factors as the supply of materials, maintenance and technical operation of facilities and equipment, unavailable equipment, or variable manufacturing processes. Any of these factors may result in the generation of nonconforming batches or in GMP- or regulatory-compliance issues being identified by the company, health authorities, or others.

As part of the EMA interassociation drug shortages project, PDA and the International Society for Pharmaceutical Engineering (ISPE) coordinated efforts at both the product and systems levels to address the prevention of drug shortages caused by manufacturing and quality problems. They advised that companies need to focus on operating effectively in an environment of increasing global complexity while continuing to supply products.

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TR-68 provides a holistic, risk-based framework at a product level for the prevention and management of drug shortages caused by manufacturing and quality issues. Risk-based approaches and risk-based decision-making are more broadly expected, accepted, and integrated into regulations and regulatory oversight mechanisms. However, the industry and regulators are still on a learning curve around the consistent application and dynamic use of risk-based approaches and risk-based decision-making in this area.

4.0 Management Responsibilities

A drug manufacturing company's interest is in developing, producing, and making pharmaceutical or biopharmaceutical medicines available to patients in a financially sustainable manner. Once approved by relevant health authorities, a drug should be available with no interruptions to supply; the manufacturer is accountable for producing and making the drug available while the wholesale distributors have a general service obligation to bring the drug to the patients. Senior management of the MAH is accountable for decisions that may impact the company's ability to cause, prevent, and/or recover from drug shortages. It is also accountable for ensuring the availability of their medicinal products in the amount needed and of the right level of quality. To ensure continued supply, it is important for management to ensure regulatory compliance, as well as compliance to GMP and good distribution practices (GDP), and a robust quality culture that is maintained and enhanced over time. Elements of this quality culture include:

- Competent employees
- Desired behaviors
- Learning environment
- Decision-making (in areas that can impact the availability of drugs, such as decisions related to new supplier approvals, quality defects, deviations, and change controls)
- Proactive QRM (i.e., ability to foresee and prevent drug shortages)
- Environments where people can speak up freely about issues
- Opportunities for improvement and interest in continuous improvements (e.g., technologies, systems, working methods)

Maintaining compliance with GMP requirements across all sites and functions is a prerequisite to ensure uninterrupted supply. Senior management of the MAH is ultimately accountable for maintaining manufacturing sites and other areas covered by pharmaceutical regulations, at a minimum, in compliance with GMPs and relevant regulatory requirements and/or commitments. This includes the MAH accepting accountability for oversight and the mitigation of any product quality and/or supply risks at third-party suppliers or service providers (e.g., contract development and manufacturing organizations (CDMO), contract laboratory organizations). Contracts and third-party relationships should be established with clear expectations and obligations related to quality and managing capacity constraints that might impact product supply or result in shortages.

Where noncompliance is identified, either by company employees via internal audits or other means or through health authority inspections, all efforts should be made to correct it and ensure compliance.

When a drug shortage is anticipated, senior management is responsible for initiating a dialog with the pertinent regulatory authorities. The objective of the dialog should be to reach a viable solution that considers the elements of patient safety, GMP/GDP and regulatory compliance, drug availability, technical abilities, and sustainable business operations.

Applied quality principles aimed at continually improving quality performance are also expected to be fully integrated into business operations and to go beyond compliance with the minimum GMP/GDP requirements.

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There are many ways in which the MAH can ensure uninterrupted supply and receive early warning of potential supply issues; this technical report provides a structured approach to achieve such an overview using a risk triage model. Companies can consider assigning a customer service representative, a role analogous to an internal auditor or risk manager, to monitor issues that may lead to drug shortages (see **Section 3.3**). Those assessments would provide management with the timely information necessary to understand the risks to patients and the business because of drug shortages. Such a responsibility would be valuable because of the diversity of causes of drug shortages.

Once management obtains information about shortages, it must develop an action plan for addressing the drug shortage risks and make risk-based decisions on how resources will be managed and, potentially, how capital is invested to ensure appropriate capacity and redundancy.

5.0 Risk-Based Approaches for Prevention of Drug Shortages

A holistic approach to the prevention and management of drug shortages focused on patient safety is important and complex. There are multiple elements, both at the system and product levels, that work individually or in combination to prevent a shortage:

- At the product level, an end-to-end understanding of the value chain for manufacturing and distribution provides the ability to understand and implement end-to-end controls.
- At the system level, appropriate oversight and a robust pharmaceutical quality system (PQS) ensure compliance with applicable laws and regulations ***(22)***.

Both approaches should also establish mechanisms that provide early warning signals of potential shortages, including near-misses, assess potential root causes for the shortage considering multiple aspects (**Figure 3.0-1**), drive targeted manufacturing and product quality improvements (e.g., introduction of new technologies, continual improvement, innovation), and enable timely communications with key stakeholders and health authorities. Other foundational elements should:

- Ensure strong management accountability
- Foster a sustainable quality culture that enables employees to speak up
- Reward behaviors that drive joint ownership between operations and quality units for timely issue resolution
- Build people skills and capabilities that drive change leadership and excellence in pharmaceutical operations
- Establish a governance model that tracks operational and quality performance with appropriate metrics ***(40)***.

As part of a robust quality system, companies should implement QRM principles and knowledge management capabilities to accomplish the following ***(1, 22)***:

- Proactively identify what controls are, and are not, in place for shortage prevention (for critical products at least)
- Proactively identify and control risks of interruption in manufacturing and product supply
- Link to clinical and safety risk assessments for their products
- Facilitate risk-based decision-making by companies and health authorities in response to shortages in order to control and reduce harm to patients

Opportunities for informed and timely decision-making based on QRM principles, knowledge management, and a good understanding of patient impact in the event of a shortage allow companies

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to proactively manage and prevent drug shortages. When drugs become unexpectedly unavailable in the manufacturing supply chain, risk-based decision-making enables companies to provide key risk-indicating information to health authorities in a timely manner before the shortage becomes severe *(38)*.

In addition to internal risk and knowledge management, it is helpful to have systems that generate reliable and up-to-date information about the marketplace for a company's products (e.g., trends in demand, usage volumes, market-share data). This combined body of information can also be very useful in decision-making for both the company and health authorities in order to respond to a shortage. It can help both parties understand what the risks of a shortage are to patients, what advice to communicate to healthcare providers, and what measures to take in order to respond quickly to a drug shortage. Information about the supply position for competitor products and alternative therapies is also important for assessing the risks of shortage-related issues, although this information may not always be readily available due to confidentiality and other considerations. This information can be used, for instance, in the event of a shortage when the supply of an alternative product, where available, can be increased to minimize the impact to patients.

Figure 5.0-1 depicts the key components of a holistic framework at the end-to-end product-value-chain level and in terms of the end-to-end product, process, system, inventory, and diversification controls for proactive prevention and management of drug shortages.

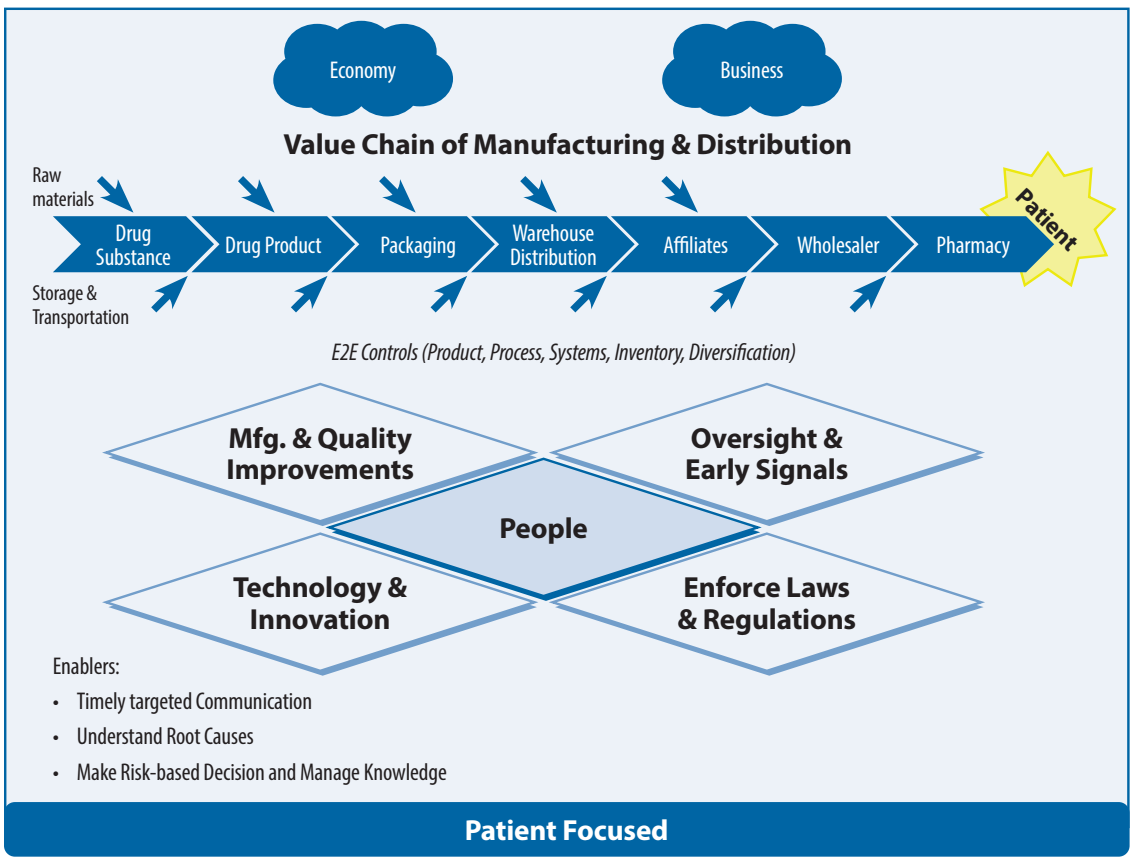

Figure 5.0-1 Holistic Framework for End-to-End Drug Shortage Prevention

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5.1 Concepts of Risk and Knowledge Management

According to ICH *Quality Guideline Q10: Pharmaceutical Quality System*, QRM and knowledge management are enablers of the PQS. As such, QRM and knowledge management are intended to facilitate the practical integration of all elements of the PQS "to enhance the quality and availability of medicines around the world in the interest of public health" ***(22)***.

A good understanding of the manufacturing process and end-to-end supply chain, up to the transfer of product from the MAH to the first economic customer (i.e., to wholesale distributors), is necessary for appropriate QRM to identify, assess, control, and review the risks of potential drug shortages. Once the hazards are identified and the risks are assessed, appropriate risk control measures can be implemented to reduce (control, mitigate, or eliminate) the risks of potential drug shortages before they lead to actual drug shortages. Of course, not all hazards can be anticipated; in such instances, as soon as new information on a hazard becomes available, risk-review activities can ensure timely assessment and implementation of risk control actions.

Knowledge management supports the PQS by facilitating easier access to product and process knowledge throughout the product's lifecycle. This availability of knowledge regarding process-performance and product-quality monitoring systems can enhance process understanding. In addition, ICH Q10 calls for process and product monitoring throughout product development and commercial manufacturing, so the information can be used to establish and actively manage a timely control strategy for manufacturing. Finally, management review of process performance and product quality should identify appropriate actions for continual improvement, such as the capture and dissemination of knowledge and the identification of post-approval changes (PAC).

Most companies have mechanisms to respond to drug shortages when they occur; however, these mechanisms might not be comprehensive or fully integrated into company procedures. A comprehensive approach for managing potential drug shortages should involve:

- Ability to respond quickly to minimize the impact of a shortage
- Framework for anticipating drug shortage triggers and risks and for establishing mechanisms to control risks and prevent shortages
- Mechanisms and processes for timely communications

The EFPIA *Good Practice Guide: Reducing Risk for Drug Product Shortages* describes a "good practice" model that a company can adopt to prevent and/or manage potential drug shortage situations ***(41)***.

A company's QRM policy and related procedures should address risks to patient safety in the context of product quality and availability. Typically, risk ranking levels in a company's QRM policy include ranking for GMP and regulatory compliance risks and patient safety risks based on product quality. **Table 5.1-1** provides an example of a risk ranking scale that incorporates considerations for both product quality and product availability in the ranking of patient safety risks.

While manufacturing and quality issues could impact product quality or patient safety, there may be instances when product quality or patient safety are not impacted. Still, there may be issues with regard to compliance with GMP or GDP or regulatory commitments provided in the product dossier. When handling GMP and regulatory compliance issues or deviations that may negatively impact the release of a product, the company must quickly identify the potential issue and work on a solution(s) to mitigate the risk of a drug shortage. Hence, it is important that the education and qualification of employees who manage deviations include aspects associated with the assessment of potential supply impact in case an issue evolves further. It is also important to establish mechanisms for escalating these GMP, GDP, and regulatory compliance issues to the appropriate level of management, as solutions might often require the allocation of resources and identification of alternatives. Companies often handle such issues internally, but they should involve the relevant health authorities early on in case there is a potential risk of a drug shortage, and a solution is unachievable in a short timeframe.

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Table 5.1-1 Examples of Risk ranking: Patient safety Ranking Incorporates Product Quality and Product Availability

Risk Ranking	Impact to Patients (Product Quality and/or Product Availability)	Impact to GMP Compliance
High	• Irreparable patient harm or death • Shortage of a medically necessary (life supporting or life sustaining) product	• Consent decree or warning letter • Withdrawal of GMP certificate or manufacturing authorization • Critical health authority observations or repeat inspection observations • Systemic breakdown of GMP systems • Recall
Medium	• Reversible patient harm • Shortage of product used for acute short-term or chronic long-term product indications	• Major health authority observations • High complaint rate
Low	• No product or patient impact • No shortage or multiple sources available for products that are not medically necessary	• Minor observations • Minor departures from GMPs

The process of knowledge management follows the QRM process according to ICH Q9(R1). Understanding and managing risks provides knowledge that can then be used to appropriately prevent and manage risks on an ongoing basis. Thus, the QRM process explained in PDA TR-54 is a foundation for the understanding and implementation of appropriate knowledge management. It is important in this context to understand which questions are answered by the knowledge available and who needs the knowledge to enable the prevention and management of risks, including the risks of drug shortages. Therefore, an understanding of quality risks based on the available knowledge about the product, manufacturing process, value chain, and overall effectiveness of the quality system forms the foundation for all efforts performed and resources spent. Without this fundamental understanding, activities related to risk and knowledge management may not be fruitful.

5.2 The Importance of Interacting with Health Authorities

There is a legal requirement for MAHs to notify relevant health authorities of any potential drug shortages in the European Union and in the United States ***(17, 18)***. Therefore, early interactions should take place between a pharmaceutical company and relevant health authorities when a meaningful disruption in supply is expected ***(38, 42)***. Laws and regulations related to drug shortages are continuing to evolve and are defining or clarifying responsibilities for stakeholders, such as industry, patient, and healthcare professional organizations, with respect to ensuring uninterrupted drug supplies to patients ***(43, 44)***.

Health authorities generally prefer to be involved in several aspects of drug shortage management and response. For example, they might want to:

- Oversee how the company or MAH is managing the shortage
- Engage other manufacturers of the same product or alternatives to increase their production
- Coordinate actions, communications, and responses to the shortage among various stakeholders, including other MAHs and manufacturers, as well as patients, healthcare professionals, and government agencies that purchase medicines
- Ensure that adequate corrective actions are taken to prevent recurrences

At a higher level, health authorities may wish to contribute to an environment in which supply chains become more robust and shortages can generally be prevented. This can support the primary goal of health authorities, which is generally the protection of public health and the availability of medicinal products.

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Health authorities usually take a risk-based approach to shortages. For example, they participate in the management of and response to shortages that may have an impact on patients now or in the near future. Health authorities, however, generally do not have the resources to become closely involved in all aspects of shortages. Risk assessments can help identify shortages that require their attention. Health authorities and industry have a mutual interest in understanding the causes of shortages, even where there is little to no impact on the patient population, to prevent them from recurring.

Knowing which shortages may or may not impact patients requires careful analysis as well as effective multistakeholder and media communications management, and the situation may vary from country to country. Robust risk assessment based on robust knowledge management of the manufacturing and supply processes and active, timely dialog with health authorities and other relevant stakeholders can facilitate this determination.

6.0 Risk Triage Model

A preventive approach to managing the risk of drug shortages caused by manufacturing or quality issues includes a risk assessment method called "triage." A triage approach allows a quick and easy assessment of the shortage risks of a product and prioritizes actions for proactive drug shortage management (risk control). Not all products require proactive drug shortage management. Only medically necessary and critical products and those with the greatest impact to patients require commensurate levels of effort and planning to ensure that the supply is adequate, and the availability component of the risk calculus is assessed and appropriately managed. The method and tools proposed in **Section 6.0** are intended to help determine whether a proactive Drug Shortage Prevention and Response Plan is required for a product.

Companies should have robust QRM and knowledge management systems to both prevent (proactive) and respond (reactive) to drug shortages (see **Section 5.0**). Proactive QRM for drug shortage prevention should involve the following:

- Categorizing each product by criticality based on its indication, marketplace importance, profile, current supply situation, and patients' needs
- Establishing and periodically reviewing a Drug Shortage Risk Register—a single source of information on hazards and the related risks that can result in drug shortages—by proactively identifying and assessing risks to quality, compliance, and supply that could lead to a shortage. A Drug Shortage Risk Register (see **Section 11.0: Appendix 1** for a sample risk register template) should provide the following information, which should be updated and reviewed on a regular basis:
    - Description of potential drug shortage risks for a product, including API, excipients, and sources of risk
    - Countries or regions that could be affected
    - Description of the current controls that are or are not in place, with respect to the prevention of shortages, and the known effectiveness of such controls
    - Risk level for a potential drug shortage, considering the current controls, new risk control actions that are required, owners, and due-date status
    - Key leading and lagging indicators that would trigger early and timely responses to potential shortages, including near-misses on a batch release or the ability to meet product demand

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- Ensuring timely reduction and management of risks by completing risk control actions defined in the risk control plan based on criticality of product availability. Issues to address in the risk control plan could include:
  - Inventory and safety stock planning
  - Demand and supply planning for all relevant markets using a sourcing strategy (e.g., dual sourcing)
  - Capacity planning for manufacturing and distribution (capacity reserve)
  - Availability of alternative therapies
  - Acceleration of health authority actions (e.g., approval of post-approval changes, inspections)
- Establishing a Drug Shortage Prevention and Response Plan, at least for medically necessary products, which could include early warning systems for potential shortages of certain products (see **Section 12.0: Appendix 2** for a sample prevention and response plan template).

Responding to and disseminating information about a potential drug shortage in a timely and effective manner is especially important for medically necessary, life supporting, or life sustaining therapies. Since some shortages need to be addressed quickly, a preestablished and well-structured drug shortage response plan can be very valuable in laying out a clear path of actions to take to minimize the impact of a shortage, end the shortage, and quickly return product supply to its normal state.

6.1 Description of the Risk Triage Model

The risk triage method uses a preliminary hazards analysis (PHA) approach, which simplifies the process after information is collected on the potential or actual drug shortage. A four-step PHA triage approach is used to evaluate the relationship of three risk elements—therapeutic use of a product, availability of alternatives, and likelihood of occurrence—to determine the risk of a drug shortage. The steps in the triage process depicted in **Figure 6.1-1** are as follows:

1. Identify impact level (impact on a patient) based on therapeutic use and availability of alternatives.
2. Determine the likelihood of a drug shortage for the product, taking into account any drug shortage prevention controls that are in place and their likely effectiveness.
3. Determine the risk priority level based on a combination of impact on the patient and the likelihood of a drug shortage for the product.
4. Plan and implement the suggested risk controls for the assessed product based on the risk priority level.

The details of the triage model displayed in **Figure 6.1.1-1** are described step by step in **Sections 6.1.1-6.1.4**.

The triage method incorporates key words used to create a simple risk assessment design approach. Rather than using very general, subjective terms to describe priority levels for each of the three elements used in the triage model (e.g., high, medium, and low severity), using discrete information allows the assessment to focus on product information rather than defining "high," "medium," and "low." **Table 6.1-1** shows how the key words are used to assign a priority level for each of the three elements in the risk triage model ***(45)***. The current Shortage Prevention Controls will be listed in the drug shortage prevention plan.

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1. Define impact to patient

Therapeutic Use & Consequences if Alternate Product not Available		Availability of Alternatives: No Alternatives Available	Alternative Products Available: Similar Therapy	Exact Product Available but in Other Presentations
Medically necessary product, life supporting or life sustaining	Fatal or severe irreversible harm if the patient is not treated with the product	Risk Level A	Risk Level A	Risk Level B
Acute short term or chronic long term	Severe harm but reversible if patient is not treated with the product	Risk Level A	Risk Level B	Risk Level C
Other indications	Inconvenience if patient is not treated with the product	Risk Level B	Risk Level C	Risk Level C

2. Assess the likelihood of shortage: at each level consider the likelihood of drug shortages and the risk controls to prevent it

3. Define the risk priority level

Therapeutic Use & Consequences if Alternate Product not Available	Likelihood of Shortage: High	Medium	Low
Risk Level A	Risk Priority Level 1	Risk Priority Level 1	Risk Priority Level 2
Risk Level B	Risk Priority Level 1	Risk Priority Level 2	Risk Priority Level 3
Risk Level C	Risk Priority Level 2	Risk Priority Level 3	Risk Priority Level 3

4. Risk Output: Develop and implement preventive risk control strategies

Risk Priority	Suggested controls/preventive actions
Level 1	Preventive actions and/or response plan required
Level 2	Preventive actions and/or response plan recommended; option to accept residual risk
Level 3	Generally accepted risk level

Figure 6.1.1-1 The Four Steps in the Risk-Based Approach to Drug Shortage Prevention

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Table 6.1-1 Key Words used in the Risk Triage and the Link to the Likelihood of Shortages

Risk Triage	Therapeutic Use of Product	Availability of Alternatives	Likelihood of Shortage
High	Medically necessary, life supporting, or life sustaining: • Used in acute or chronic situations of stable conditions or disease with a fatal outcome where the product has been shown to affect the progression of the disease or survival	No alternatives available	High likelihood of shortage
Medium	Acute short-term or chronic long-term: • Part of a treatment for or prevention of a disease that is life-threatening or irreversibly progressive, or without which public health could be severely harmed	Alternative products or APIs available, e.g.; • Alternative dosing (lower dose/ temporary break from treatment) or limiting the use to high risk patients could be explored; consider this might depend on the expected duration • Other products in the same class or even other classes, consider adverse events, or narrow therapeutic range or contra-indications	Medium likelihood of shortage
Low	Other indications	Same product or API available, e.g.: • Sourced from an alternative manufacturing site • Consider manufacturing capacity and technical and regulatory times to switch • Different strengths or formulations; consider need for formulations suitable for use in special populations	Low likelihood of shortage

6.1.1 Step 1: Define Impact to the Patient

Companies can benefit from categorizing their products by therapeutic use, which makes it possible to apply a risk-based approach for managing drug shortage risks for their portfolio of products. A risk level indicating the severity of the impact of a shortage on the patient is assigned based on two factors: therapeutic use of the product and availability of alternatives, as shown in **Table 6.1.1-1**.

Table 6.1.1-1 Step 1: Define Impact on Patient

1. Define impact to patient			Availability of Alternatives		
			No Alternatives Available	Alternative Products / API Available	Same Product / API Available
Therapeutic Use & Consequences if Product not Available	Medically necessary product, life supporting or life sustaining	Fatal or severe irreversible harm if the patient is not treated with the product	Risk Level A	Risk Level A	Risk Level B
	Acute short term or chronic long term	Severe harm but reversible if patient is not treated with the product	Risk Level A	Risk Level B	Risk Level C
	Other indications	Inconvenience if patient is not treated with the product	Risk Level B	Risk Level C	Risk Level C

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"Medically necessary products," "critical medicines," and "life supporting or life sustaining products" have the highest criticality based on patient need; therefore, their shortages present the greatest risk. (Definitions of terms are provided in **Section 3.0**.) Ensuring an uninterrupted, reliable supply of these products at all times to all patients who need them is key; therefore, robust controls must be in place to proactively manage the risk of shortages of these drugs.

Products used to treat acute short-term conditions or manage chronic long-term diseases (e.g., Parkinson's disease, diabetes, multiple sclerosis) constitute the next level of product indication-based criticality. Although the criticality of these products is not at the same level as that of life supporting or life sustaining therapies, the unavailability of these products over an extended period could harm patients; therefore, shortages of these products cannot be tolerated for long periods.

The lowest level of product indication-based criticality is assigned to medicinal products used for other minor indications (e.g., self-care therapies for minor ailments such as cough and cold). If a product is registered for more than one indication with different levels of indication-based criticality, the more conservative or higher level of criticality should generally be applied for this assessment. For example, if one of the product indications classifies a product as medically necessary, and the second indication classifies it for treatment of a chronic long-term condition, risk planning should be done assuming the higher level of product indication-based criticality, that is, for its use as a medically necessary product.

In addition to the therapeutic use of the product, the second dimension in ranking the impact of a shortage to patients is access to and availability of alternative therapies. Relevant health authorities may need to be consulted to determine where alternatives may exist for a product and for what markets they are available. Where no alternatives are available, especially for products that are medically necessary, life supporting, life sustaining, or used to treat acute short-term or chronic long-term conditions, the impact to the patient is the highest (Risk Level A – red). Therefore, the margin for tolerance of a drug shortage with these products is minimal, and controls for these risks should ensure that:

- Focus is on proactive prevention of a shortage
- Time to remedy and return supply conditions toward normal is as short as possible in the event of a shortage

Where an alternative therapy is available in the form of a similar product or other presentations of the same product, the impact to the patient is lower (Risk Level B – yellow). In the event of a shortage, it is important to ensure that these alternatives are accessible to patients in a timely manner to minimize impact (e.g., by preventing interruptions in treatment).

The lowest level of impact (Risk Level C – green) is assigned to medicinal products used for other indications (e.g., self-care therapies for minor ailments such as cough and cold) for which alternatives are readily available or for products used to treat acute short-term or chronic-long terms conditions that have the same product readily available in other presentations.

Table 6.1.1-2 provides examples of risk level assignments to products based on their therapeutic use and the availability of alternative therapies. These examples are presented only for illustrative purposes; actual rankings may vary.

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Table 6.1.1-2 Examples of Risk Levels Assigned to Products Based on Therapeutic Use and Availability of Alternatives

Product	Therapeutic Use	Consequences if Product not Available	Availability of Alternatives	Impact Level
Targeted biologics for oncology treatment (e.g., Trastuzumab)	Medically necessary, life supporting, or life sustaining product	Death or severe irreversible harm	No alternatives available	Risk Level A
Antibiotics (e.g., Azithromycin or Ciprofloxacin)	Medically necessary, life supporting, or life sustaining product	Death or severe irreversible harm	Alternative products available: similar therapy	Risk Level A
Diabetes treatment (e.g., Insulin)	Medically necessary, life supporting, or life sustaining product	Death or severe irreversible harm	Alternative products available: similar therapy	Risk Level A
Congestive heart failure medicine (e.g., Digoxin)	Medically necessary, life supporting, or life sustaining product	Death or severe irreversible harm	Exact product available but in other presentations	Risk Level B
Epilepsy medicine (e.g., Diazepam)	Acute short-term or chronic long-term condition	Severe, reversible harm	Alternative products available: similar therapy	Risk Level B
Plasma protein therapies	Acute short-term or chronic long-term condition	Severe, reversible harm	Exact product available but in other presentations	Risk Level C
OTC cough suppressants (e.g., cough syrup)	Minor indications	Inconvenience	Alternative products available: similar therapy	Risk Level C

6.1.2 Step 2: Assess the Likelihood of Shortage

In the second step of the triage, the likelihood of shortage for a product is assessed. Hazards that could increase the likelihood of a shortage should be assessed and current controls should be assessed to identify the hazards. Where good, robust controls exist for a hazard, the likelihood of occurrence would be low; if there are limited to no controls for a hazard, the likelihood of occurrence increases. Historical data may be available for certain hazards, and this historical data should be leveraged to assess the relative likelihood of risks. For example, product sourcing may be one of the controls assessed to determine the likelihood of a shortage. Single-sourcing of a product or key material supplier increases the likelihood of occurrence of a shortage in the event of an issue at that sole-sourced facility or supplier. Other controls may be related to infrastructure maintenance and improvements, technology updates, business continuity planning and capacity assessment, process capability, inventory levels, or supplier diversification.

This assessment of likelihood of occurrence may not be quantitative, which is considered acceptable because it is more important to assess the controls and their likely effectiveness and to understand the relative likelihood of the occurrence of a shortage for identified hazards. In this risk triage model, a simple three-level (high, medium, low) scale for assessing the likelihood of occurrence has been proposed.

6.1.3 Step 3: Assess Risk Priority Levels

In Step 3, outputs from Step1 (i.e., Risk Level A, B, or C), in combination with the likelihood of shortage, are used to assign a Risk Priority Level (1, 2, or 3) as shown in **Table 6.1.3-1**.

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Table 6.1.3-1 Assess Likelihood of Shortages (Step 2) and Risk Priority Levels (Step 3)

Process Steps: 1. Take the impact to patient (Risk level A, B or C) 2. At each risk level, consider the likelihood of drug shortage (Step 2) 3. Define priority (Step 3)		Likelihood of Shortage		
		High	Medium	Low
Therapeutic Use & Consequences if Alternate Product not Available	Risk Level A	Risk Priority Level 1	Risk Priority Level 1	Risk Priority Level 2
	Risk Level B	Risk Priority Level 1	Risk Priority Level 2	Risk Priority Level 3
	Risk Level C	Risk Priority Level 2	Risk Priority Level 3	Risk Priority Level 3

A single-sourced product with no backup or a product with insufficient safety stock presents the highest risk and, in combination with Risk Levels A and B (from Step 1 of the triage), would result in the highest risk priority level (Risk Priority Level 1). A multisourced product with a low capacity reserve (e.g., preparation of a backup manufacturing facility to produce the product will take at least six months) or with sufficient short-term inventory has a medium likelihood of occurrence and, in combination with Risk Levels A and B (from Step 1 of the triage), would result in a medium risk priority level (Risk Priority Level 2). A multisourced product with an adequate capacity reserve (e.g., an active backup manufacturing facility or will require less than four months to prepare to produce the product) and a sufficient inventory level represents the lowest risk and, in combination with Risk Levels B and C, would result in the lowest risk priority level (Risk Priority Level 3).

A simple example of an assessment is shown in **Figure 6.1.3-1**. In this example, no alternative therapies are currently marketed or are in clinical trials for a biological drug product used to treat late-stage cancer. The product is dual-sourced but does not have enough inventory reserves to cover a major manufacturing or quality issue.

These risk ratings are used in the two-step triage risk analysis shown in **Figure 6.1.3-1**. The high risk categorization for therapeutic use and availability of potential alternatives results in designating the risk level as high (Risk Level A) from Step 1 of the triage. This likelihood of a shortage is then assessed as medium in Step 2 of the triage. In Step 3, Risk Level A combined with the medium likelihood of a drug shortage results in the overall Risk Priority Level 1. Given this risk priority level, a Drug Shortage Prevention and Response Plan needs to be developed for this product (see **Section 7.0**).

6.1.4 Step 4: Develop and Implement Preventive Risk Control Strategies

After the risk priority levels are assigned to the product in Steps 1-3 of the triage, appropriate risk control strategies (may be predetermined in some instances) should be developed and implemented.

Some actions to consider based on the risk priority level are:

- Risk Priority Level 1
  - Develop a value stream map (VSM) (starting materials through final drug product packaging and distribution) to identify and reduce inventory and manufacturing risk stress points. Identify appropriate inventory levels for each manufacturing stage based on the current state of controls, probability of drug shortage, and how long a shortage can be tolerated without adverse effects on patients. In addition, the need for a multisourcing strategy should be assessed, taking business considerations into account with senior management involvement.

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Define impact to patient

Therapeutic Use & Consequences if Alternate Product not Available			Availability of Alternatives: No Alternatives Available	Alternative Products Available: Similar Therapy	Exact Product Available but in Other Presentations
	Medically Necessary Product, Life supporting or Life sustaining	Fatal or severe irreversible harm if the patient is not treated with the product	Risk Level A	Risk Level A	Risk Level B
	Acute short term or chronic long term	Severe harm but reversible if patient is not treated with the product	Risk Level A	Risk Level B	Risk Level C
	Other indications	Inconvenience if patient is not treated with the product	Risk Level B	Risk Level C	Risk Level C

Process Steps:

1. Take the impact to patient (Risk level A, B or C; here A)
2. At each risk level, consider the likelihood of drug shortage (Step 2)
3. Define priority (Step 3)

Therapeutic Use & Consequences if Alternate Product not Available	Likelihood of Shortage: High	Medium	Low
Risk Level A	Risk Priority Level 1	Risk Priority Level 1	Risk Priority Level 2
Risk Level B	Risk Priority Level 1	Risk Priority Level 2	Risk Priority Level 3
Risk Level C	Risk Priority Level 2	Risk Priority Level 3	Risk Priority Level 3

Define suggested controls/preventive measures as of the triage output

Risk Priority	Suggested controls/preventive actions
Level 1	Preventive actions and/or response plan required
Level 2	Preventive actions and/or response plan recommended; option to accept residual risk
Level 3	Generally accepted risk level

Figure 6.1.3-1 Example of Applying the Risk Triage Model towards the Drug Shortage Prevention and Response Plan

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- — Review supplier management controls for high risk products and high risk suppliers of the components of these high risk products. Implement controls for these products where gaps are identified in supplier management controls.

- — Proactively create and maintain a plan to communicate shortage risks with health authorities, distribution channels, and users. The guide *Quality and Manufacturing Driven Supply Disruptions: Industry Communication to Authorities*, developed by the European industry collaboration of EFPIA, EGA, AESGP, and PPTA, provides principles and a framework to harmonize reporting and communications between MAHs and health authorities about drug shortages due to quality and/or manufacturing disruptions ***(38)***.

- — Implement an effective Stage 3 continuous process verification (CPV) or ongoing process-verification monitoring program for CPV data that could help identify manufacturing issues before they occur.

- Risk Priority Level 2
  - — Consider additional proactive inventory management controls and a multisourcing cost–benefit analysis.

  - — Consider VSM for specific manufacturing processes to identify inventory stress points and potential manufacturing problems.

  - — At a minimum, ensure that CPV plans for these products to undergo additional review as part of the company's drug shortage management activities.

- Risk Priority Level 3
  - — A Drug Shortage Prevention and Response Plan is not necessary, but these risks should be reviewed regularly.

High and medium risks, along with the risk control actions and level of residual risk in the Drug Shortage Risk Register, should be tracked, periodically reviewed, and managed.

Examples of risk control strategies are provided in **Table 6.1.4.** The level of effort and formality required for each product assessment depends on the level of risk identified. These examples are not all-inclusive. Organizations should evaluate and determine which strategies for drug shortage control will be the most effective for the risks identified for their specific product and its use. Where appropriate, these strategies should be documented in procedures, integrated into the quality system and/or business processes, implemented in a consistent manner across the company, and actively managed using a Drug Shortage Risk Register. **Section 11.0: Appendix 1** provides a template for creating a Drug Shortage Risk Register.

In general, risk control strategies vary, based on product specifics or its node in the supply chain, for example. The listed factors and stakeholders in the root causes for shortages seen in **Figure 3.0-1** should also be considered.

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Table 6.1.4 Examples of Areas to Assess for Potential Risk Control Measures (not all-inclusive)

Area of Focus	Examples of Areas to Assess for Potential Risk control Measures
Infrastructure and QMS Controls	• API and medicinal product production (material receipt and testing, dispensing, bulk product manufacturing, packaging, quality control, release) • Storage and logistics locations (e.g., near to key transport hubs) • Manufacturing facility and equipment • Maintenance • Aging facilities • Management priorities • Personnel resources • Time for external quality assurance testing as appropriate (e.g., Official Medicines Control Laboratories, import testing) • Qualification and validation • Production planning
Technology	• Innovative manufacturing (including digitalization) and testing • Emerging technology • Outdated technology • Device sourcing and assembly when highly fragmented across different suppliers and manufacturers
Materials and Supplier Management	• Material sourcing • Alternate or backup site (including redundancy) • Inventory and safety stock management • Value stream mapping • Demand forecasting • Diversification, multisite sourcing with higher manufacturing capacity reserves • Supplier qualification and ongoing supplier management and monitoring • Supply chain or transportation line security • Supplier relationship management
Business Continuity	• Understanding of marketplace dynamics • Geopolitical considerations • Worst-case scenario planning • Disaster planning (e.g., natural disaster, climate impact, pandemic, significant supply-chain disruptions) • Production capacity • Recovery plan • Reliance on domestic or foreign manufacturing and supply • Personnel management in case of crises (e.g., public health emergencies e.g., pandemic, strikes)
Regulatory Considerations	• Regulatory requirements and expectations • Post-approval changes • Evolving regulatory landscape

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6.2 Supply Chain and Value Stream Mapping

The MAH is accountable for ensuring that risks for patient safety, product quality, and product availability are adequately managed in the end-to-end product supply chain. In addition to assessing the risks with various suppliers and external service providers (e.g., CDMOs, CLOs) and addressing them through their supplier management controls, the MAH should also review the QRM processes used, especially by their key suppliers and service providers. This can help ensure that these suppliers and service providers have an adequate framework in place, in accordance with the principles of ICH Q9(R1), to assess and manage their quality and compliance risks.

Process mapping, in general, can play a key role in managing the risk of drug shortages particularly as supply chains continue to become increasingly complex. Mapping of the end-to-end product supply processes is one of the most important tools that can be leveraged for managing the risk of drug shortages due to manufacturing and supply problems. VSM (also known as material and information flow mapping), a particularly powerful tool, graphically displays the flow of the manufacturing and transportation processes for a product and highlights potential manufacturing, inventory, and supply stress points. Although VSM is often used to reduce amounts of inventory and inefficiencies across the supply chain, it is also used to maintain proper levels of in-process inventory to ensure predefined final product inventories. Maintaining agreed-upon inventory levels of products required for pandemics or flu seasons is an example of how VSM can be used. **Figure 6.2-1** displays an example of how VSM serves as a useful tool to help manage drug shortages proactively and reactively.

While VSM is a great tool for mapping products and information flow, it is not the only approach available. Other tools that could be used are DRIVE (Define, Review, Identify, Verify, Execute), ICOR (Inputs, Outputs, Controls, and Resources), Cause-and-Effect Diagrams, and Process Flowcharting.

The following elements constitute an effective mapping tool:

- **Team Makeup:** The mapping team should be cross-functional and include individuals who are intimately familiar with manufacturing and supply processes and equipment. The most successful mapping exercises involve laboratory technicians and operators. Mapping requires good facilitation, preferably by an objective, neutral individual.

- **Planning and Preparation:** The scope of the mapping (local to a site or region or broader in the supply chain) should be identified in advance. The team should establish timelines and goals for the mapping (in this case, specific to drug shortages). The exercise will likely become part of a larger initiative for developing a Drug Shortage Prevention and Response Plan.

- **Evaluation:** The team should perform all steps of the mapping process and capture data and information on how the process works. The team should also have access to existing procedures as well as quality events to capture all stress points. Depending on the mapping technique used, specific symbols are used to indicate times, inventory, issues, and areas for improvement.

- **Map:** After a map of the current state is drawn, listing all the steps and handoffs in the process, a future-state map can be drawn using the same steps to identify risk reducing actions that will meet the needs of the product.

- **Action List:** To develop and maintain an effective Drug Shortage Prevention and Response Plan, an action list needs to be developed to address the gaps and risks based on the future-state map actions.

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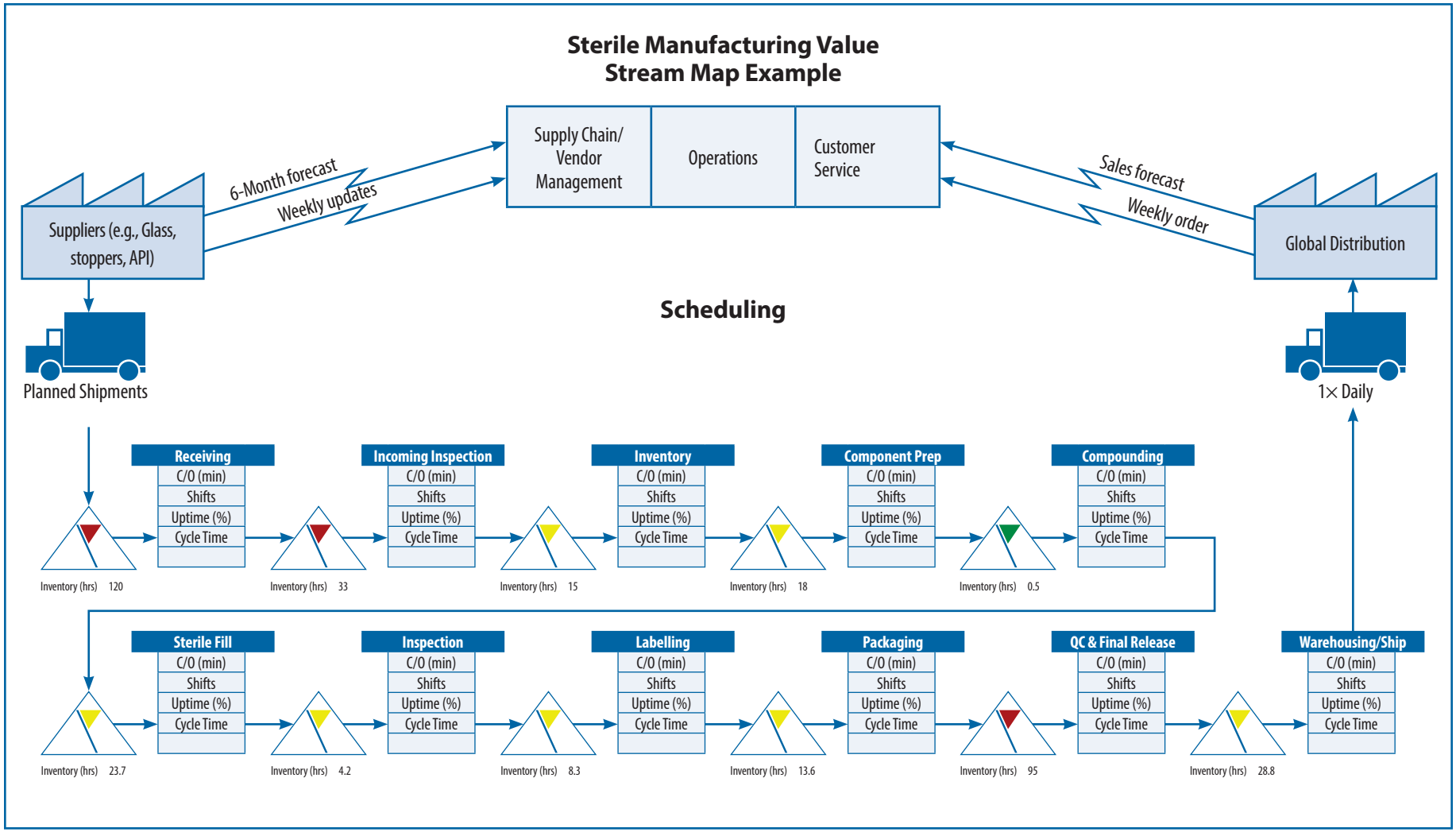

Figure 6.2-1 Example of a Typical Value Stream Map

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6.3 Managing Drug Shortage Risks Due to Aging Manufacturing Facilities and Technologies

Manufacturing and quality risks introduced by aging or legacy facilities and technologies (e.g., out-of-date manufacturing facilities, equipment, analytical technologies) that are not fully up to current standards or expectations can contribute to an increased likelihood of issues that could potentially result in drug shortages. These risks should therefore be proactively identified and managed during the lifecycle of the manufacturing facility, process, equipment, or analytical methods. In order to accomplish the objectives of ICH Q10, continual improvement is an important element of product realization and of maintaining a state of control. The EU GMP guidelines on pharmaceutical quality systems expect for medicinal products that:

> Product realization is achieved by designing, planning, implementing, maintaining, and continuously improving a system that allows the consistent delivery of products with appropriate quality attributes. Continual improvement is facilitated through the implementation of quality improvements appropriate to the current level of process and product knowledge ***(46)***.

ICH Q9(R1) states that a robust facility infrastructure can facilitate reliable supply; it includes suitable equipment and well-designed facilities for manufacturing (including packaging and testing). Multiple factors can affect robustness, such as an aging facility, insufficient maintenance, or an operational design that is vulnerable to human error. Risks to supply can be reduced by addressing these factors, as well as through the use of innovative technology, such as digitalization, automation, and isolation technology, among others.

Updating legacy facilities and technologies or implementing new innovative technologies can present multiple challenges, not always cost-related, that prevent companies from pursuing these strategies as part of lifecycle management. However, where aging facilities or technologies increase the risks to product quality, patient safety, and/or supply interruption, it becomes essential for manufacturers to address these risks.

6.3.1 Causes and Effects of Aging Facilities, Processes, and Analytical Technologies

Manufacturing facilities, processes, and analytical technologies age and, over time, may no longer be capable of meeting performance requirements or delivering the expected product quality attributes. Some of the main reasons why facilities, processes, or analytical technologies become obsolete are:

- Lack of long-term planning and poor understanding of the importance of reinvestment and recapitalization of facilities and processes
- Concern about the unanticipated negative impact of improvements due to a lack of robust process knowledge or of knowledge and experience with the new technology
- Short-term focus on maintaining or reducing costs, either cost of goods sold or capital expenses
- Negative financial return on investment due to low drug prices (e.g., generics)
- Challenges, long lead times, and a lack of harmonized processes for global regulatory approvals across multiple health authorities
- Absence of positive mindset, attitude, and culture for change and continuous improvement

These factors can produce the decline that occurs over time, for example:

- As a cost-saving effort, a company may decide to move to a limited maintenance schedule on equipment or reduce the availability of critical spare parts and stock of intermediates. The lowered or partial maintenance is believed to save costs; however, breakdowns due to the lack of maintenance or a negative trend in the manufacturing environment may stop production or lead to product rejections in the future, at a far higher cost.

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- A conscious decision to drastically reduce or eliminate facility, process, and/or equipment optimization and improvement activities, over the long run, can result in antiquated and noncompliant facilities and processes with high levels of variability and risk. Additionally, aging facilities may limit the ability to increase capacity to meet the product demand.

- Maintaining a false assumption that equipment will not become obsolete in the future and will continue to be supported by the equipment manufacturer is imprudent. In almost all cases, equipment is designed with an intended life span. As equipment manufacturers move to newer models, they often identify a point in time after which they will no longer support older models. The subcontractor should be responsible for informing the equipment user as early as possible about any changes in the technology status, including obsolescence of equipment, availability of next-generation equipment, and spare parts status, so the user can change or update the relevant equipment before shortages arise. When the manufacturer fails to plan for equipment obsolescence, it often results in a quick upgrade done at a high cost and with some loss of production time.

- In some cases, a process or analytical technique was designed for one product or output, and a change in a product or output might be incompatible with the old process or product.

Quality and GMP compliance gaps associated with such aging facilities, processes, and technologies should be addressed even though the existing equipment or methods may be functional and work smoothly without problems. Well-trained personnel with a process ownership mentality may make such facilities, processes, and analytical methods run, but when such personnel retire or move to other tasks, the facility, processes, or analytical methods see their first excursions. These events occur especially when the succession personnel are not trained appropriately or rely on the existing process without knowing its weaknesses. It is very important to invest in training qualified personnel to run these often manual, processes, equipment, and methods.

In addition to not meeting current regulatory expectations, the effects of aging facilities, processes, and analytical technologies can be detrimental and may result in a total shutdown of the production of a particular drug product, potentially resulting in drug shortages. The continual maintenance and improvement of facilities, processes, and analytical methods is essential to maintain specified product quality and to assure supplies of the product.

6.3.2 Continual Improvement of Aging Facilities and Processes

Processing technology and facility designs are forever evolving. The motivation of suppliers to develop and distribute new technologies is mainly driven by industry seeking new solutions for current processing problems. However, implementation of these solutions becomes challenging in spite of the value they can bring.

Given this, implementation of new technology has resulted not just in financial viability, but an overall improvement in quality and process efficiency. For example, single-use process technology has many advantages over traditional stainless steel equipment, such as faster equipment turnaround, which enhances production capacities. Also, single-use components or unit operations can be connected via aseptic connector or tube welders, which creates a closed system that enhances the safety of both the product and the operator. The technological improvements of single-use technologies have been widely embraced, not only for new, but existing processes, as the advantages of using this technology outweigh any challenges change commonly bring. [More information about single-use systems can be found in PDA *Technical Report No. 66: Application of Single-Use Systems in Pharmaceutical Manufacturing*.]

Facility and process improvements due to technology innovation are not always as dramatic as single-use process technology or new modular cleanroom designs; they can be gradual improvements of existing equipment to the next generation. Subtle, but continual improvements demonstrate the right quality

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attitude and ownership. Such continual improvement activities inevitably lower the risk of process or facility failure, leading to operations that run more reliably and efficiently and resulting in cost reduction.

6.3.3 Lifecycle Management and Continual Improvement of Analytical Technologies

During the commercial lifecycle of a product, analytical technologies continue to improve. However, commercial product analytical control systems—in-process, release, and stability test methods, specifications and acceptance criteria, and reference-standard controls—tend to improve minimally, if at all. This slow pace of change can be attributed to such factors as high cost, resource requirements, and the complexity of the global regulatory environment for making PACs. Revisions to control systems can take two to five years, depending on their extent and the number of countries that need to approve the PAC. Revisions can take even longer for complex molecules for which physicochemical characterization data are limited, development of new methods is challenging, and new methods require new equipment in laboratories. Even if a new method is developed, collecting data to establish acceptance criteria and obtaining regulatory approvals from multiple health authorities for PACs, can take time.

A proactive approach to control system lifecycle management is important to ensure alignment with contemporary product knowledge and industry standards. In addition, companies manufacturing follow-on biologics, for example, are publishing detailed physicochemical characterizations of innovator products using new or improved technologies. Lifecycle management of analytical control systems should be reviewed periodically (e.g., every five years), not only in response to control system issues identified (e.g., through investigations), during routine commercial use.

Ensuring that the analytical control system addresses relevant product quality attributes and that these product quality attributes are well controlled throughout the product's commercial lifespan is important. Periodic assessments of an analytical control system should determine whether the system continues to control drug substance and product quality at the product's release and at the end of its shelf life. Such determinations should be based on comprehensive reviews of current information about the molecule, including adequate monitoring of relevant product quality attributes, current manufacturing process capability, alignment of acceptance criteria with process capability, compliance with evolving regulatory requirements and expectations, the ability to fulfill its intended purpose, and current analytical technologies.

Periodic assessments of the control system should include evaluations of gaps or improvement opportunities, for example, when a new critical quality attribute is identified because of new information on a molecule, such as antibody-dependent cell-mediated cytotoxicity, when superior technology is available, or when new regulatory requirements are implemented. Such gaps and improvement opportunities should be used to determine whether new methods are needed or if current methods need to be revised or eliminated. Some examples of when an update to the control system may be needed are when:

- The current method has a high failure rate, low throughput rate, or analyst ergonomic or safety limitations
- The method is no longer suitable for the control system
- The current method validation is no longer applicable
- Superior technology is available
- New or revised specifications or acceptance criteria need to be implemented

Acceptance criteria for current control system test methods should be evaluated against manufacturing and process capability (via analysis of lot-release data) and stability data (via annual stability and qualification-lot data). The process capability index measures the extent to which the process operates within its specification limits, given process variability (see *PDA Technical Report No. 59 (Revised 2012): Utilization of Statistical Methods for Production Monitoring*). A larger index indicates an increased likelihood that a result will be within the specified limits.

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Release acceptance criteria can be evaluated by comparing 95 or 99 tolerance intervals of historical batches to current acceptance criteria. If the acceptance criteria do not match manufacturing capability, those criteria may require revision. Drug substance and drug product shelf-life acceptance criteria can be evaluated by analyzing historical trend plots.

Revisions to current analytical control systems (test methods, acceptance criteria, or shelf-life specifications) require submission of an application to the relevant health authorities. Therefore, they should be coordinated as much as possible with manufacturing process improvements, transfers, and other activities requiring regulatory approval. Early discussions with relevant health authorities regarding reporting level and documentation requirements for anticipated PACs well before these changes occur will allow both the health authorities and the company to plan for the changes in a manner that will help reduce the risk of drug shortages.

6.4 Reducing Drug Shortages Through Expedited Post-Approval Changes

When companies face a potential drug shortage, they can mitigate the risk by expediting a change that may have already been submitted to a national health authority for approval. For example, a company is experiencing an unexpected increase in demand in Country 1 that cannot be met with the current capacity of Facility A. If Facility B has been approved for manufacture of the product by multiple countries and it meets the GMP expectations of Country 1, though Country 1 has not yet approved Facility B, Facility B can produce additional products using the same manufacturing process. In this example, requesting an expedited review and approval of the PAC application for Facility B in Country 1 can be a viable alternative to mitigate the risk of the drug shortage. **Section 6.4** describes how PACs can be expedited while demonstrating that patients are not exposed to any added risk because of the change.

Current regulatory PAC processes vary among countries in the levels and terminology of change categories, requirements for documentation of changes, the need for preapproval facility inspections, and timelines for approval. For these reasons, companies usually need to run multiple versions of a manufacturing process in parallel for the same product, depending on when each national health authority is likely to approve a change. Doing so is logistically challenging from the standpoint of production, inventory management, and supply. In some countries, when the anticipated PAC approval timelines do not match the forecasted demand, it can lead to the lack of availability of the product. This situation is especially challenging for biological products due to the highly complicated manufacturing processes involved; it can become even more complex when multiple changes pertaining to a product are under review at the same time by different countries.

In such cases, the company should communicate with the health authorities responsible for trade where a potential drug shortage might occur and request an expedited review of pending PACs or offer a previous version of the product if available in the inventory. Health authorities assess PACs using science and risk-based approaches while ensuring that they meet national legislative and regulatory requirements. The outcome of these assessments, ideally, should be the same regardless of the geographic location of the assessor; it almost always is. However, it would be beneficial if the regulatory approval processes for PACs could be harmonized. Harmonization would also facilitate continual improvement and product realization, two objectives of ICH Q10.

An example of an effort towards harmonization of regulatory expectations is The International Coalition of Medicines Regulatory Authorities' (ICMRA) Pharmaceutical Quality Knowledge Management System Initiative (PQKMS) ***(47)***. According to ICH Q10, Annex 1, companies that demonstrate an effective PQS and process and product understanding, including the use of QRM principles, may be eligible for enhanced regulatory approaches. This is sometimes referred to as "regulatory flexibility," where there may be an opportunity to "optimize science and risk-based post-approval change processes to maximize benefits from innovation and continual improvement" ***(22)***. Additionally, ICH *Quality Guideline*

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Q12: Technical and Regulatory Considerations for Pharmaceutical Product Lifecycle Management provides a framework via PQS to facilitate the management of PACs in a more predictable and efficient manner throughout the product lifecycle. Such opportunities may include managing more PACs in the PQS only, rather than as a prior-approval PAC. The intent of the guideline is to promote continual improvement and innovation, while strengthening quality assurance and reliable product supply ***(48)***. The Industry One-Voice-of-Quality for PAC (1VQ for PAC) initiative has developed solutions and guidance on how a company can demonstrate an effective PQS ***(49, 50)***. The PIC/S paper, *Recommendation: How to Evaluate and Demonstrate the Effectiveness of a Pharmaceutical Quality System in relation to Risk-Based Change Management,* may facilitate a health authority's assessment of a company's change management system and has the potential to provide transparency to how effectively PACs are assessed within that change management system ***(51)***.The concept has been proposed that health authorities provide incentives to companies for innovation and implementation of new technologies that reduce risk and lead to measurable shortage prevention. Some examples of innovation that health authorities might encourage with incentives are the replacement of conventional open processes with isolator technology, the use of closed rather than open manufacturing processes, enhancements in microbial and adventitious agent controls, introduction of rapid microbiology methods, and the implementation of technologies that reduce manufacturing cycle times, improve process capability, make a process more efficient, or enhance a product's analytical capabilities.

Companies and health authorities could work together to enable, where appropriate, streamlined review and approval of the following changes, provided that agreed-upon acceptance criteria from health authorities are met and that these changes do not introduce increased risks to patient safety or product quality or efficacy:

- Enhanced redundancy
  - Additional manufacturing site or line already approved in at least one significant market or country that can serve as a reference
  - Additional raw material or and API suppliers
  - Additional suppliers of container closure or other processing components
  - Additional dosage forms already approved in at least one reference country

- Improved process and product controls
  - New or improved equipment and technology
  - Manufacturing changes that lead to improved process capability
  - New analytical technologies that enhance product controls, including adventitious agent controls

- Other types of changes (provided that validated and preset criteria are met)
  - Change in reference standards
  - Use of new working cell banks
  - Addition of a product to a multiproduct facility
  - Scale-up of a manufacturing process
  - Extension of shelf life with supporting documentation
  - Change in utilities (e.g., HVAC, clean steam) or their operation with no impact on product quality
  - Change in equipment (design and/or operating principle) with no impact on product quality
  - Change in steam-in-place or clean-in-place process
  - Change in environmental monitoring program (e.g., sample location and frequency)
  - Change in cleaning methods

In all cases, the company concerned should be able to demonstrate that it has an effective PQS in place. In this regard, companies should expand their PQS to address PAC planning and assessment, especially how to conduct risk assessments and how to demonstrate risk control. The industry should work with health authorities to develop predefined global change protocols (gCP) that can be available

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and used as a standard (non-product-specific) template for the categorization, predefined data requirements, and implementation of the different types of changes. Health authorities have provided guidelines on procedures and data requirements for PACs, such as Health Canada's *Guidance Document: Post-Notice of Compliance (NOC) Changes: Quality Document* and WHO's *Annex 3: Guidelines on procedures and data requirements for changes to approved biotherapeutic products (TRS No. 1011)*. Common gCPs that are developed in partnership with health authorities for the different types of changes can help standardize activities and requirements across the industry for a particular change type such that expectations are clearly known prior to considering the change. This would help expedite the review and approval of the PAC. Such gCPs could potentially result in a lower regulatory reporting category and expedited review from multiple health authorities yet address individual health authority requirements. The gCP should include validation activities and even acceptance criteria (where possible) that should be expected for the different types of changes.

Each company could then use the gCP for the relevant type of change as a starting point in developing the submission for each of the changes in the PAC Management Protocol for their product. The company should update the predefined gCP with the specific aspects of the change for their particular product. The change(s) should not have a negative impact on product quality, safety, or efficacy. The company should describe and demonstrate through the gCP that the change does not present an increased risk to patients. Some of the changes listed in this section can be associated with potential high risks. Therefore, it is important that the risks associated with the change(s) are adequately identified, assessed, and controlled. The risk assessment for the change(s) should be included in the submission for the change to demonstrate that appropriate risk control measures are in place and provide assurance that patient safety, product quality, and product efficacy are not impacted.

7.0 Drug Shortage Prevention and Response Plan

The risk assessment triage results in an assignment of a drug shortage risk priority level to each product based on the product's indication, availability of alternative therapies, and probability of a shortage. **Section 7.0** describes how a Drug Shortage Prevention and Response Plan should be developed based on the outcomes of the risk triage. While agencies are working on new guidances, such a plan might also be known as a Shortage Prevention Plan or Mitigation Plan in the future.

A Drug Shortage Prevention and Response Plan can provide a holistic framework at a product level to address both the proactive and reactive management of a drug shortage. This plan should be proactively developed for medically necessary, life supporting, or life sustaining products and shared with health authorities during inspections. Medium risk products used for short-term acute conditions or long-term chronic conditions can also benefit from a Drug Shortage Prevention and Response Plan, especially if the risk triage has identified potential drug shortage risks at Risk Priority Level 1.

A Drug Shortage Prevention and Response Plan can be an effective mechanism for managing a product's continuously changing drug shortage risk profile. Drug shortages may present potential opportunities for product diversion and counterfeiting. Therefore, a company's surveillance program can leverage the Drug Shortage Risk Register as useful input into its process for managing product diversion and counterfeiting risks that may arise due to shortages. **Figure 7.0-1** depicts the steps involved in developing and maintaining a Drug Shortage Prevention and Response Plan.

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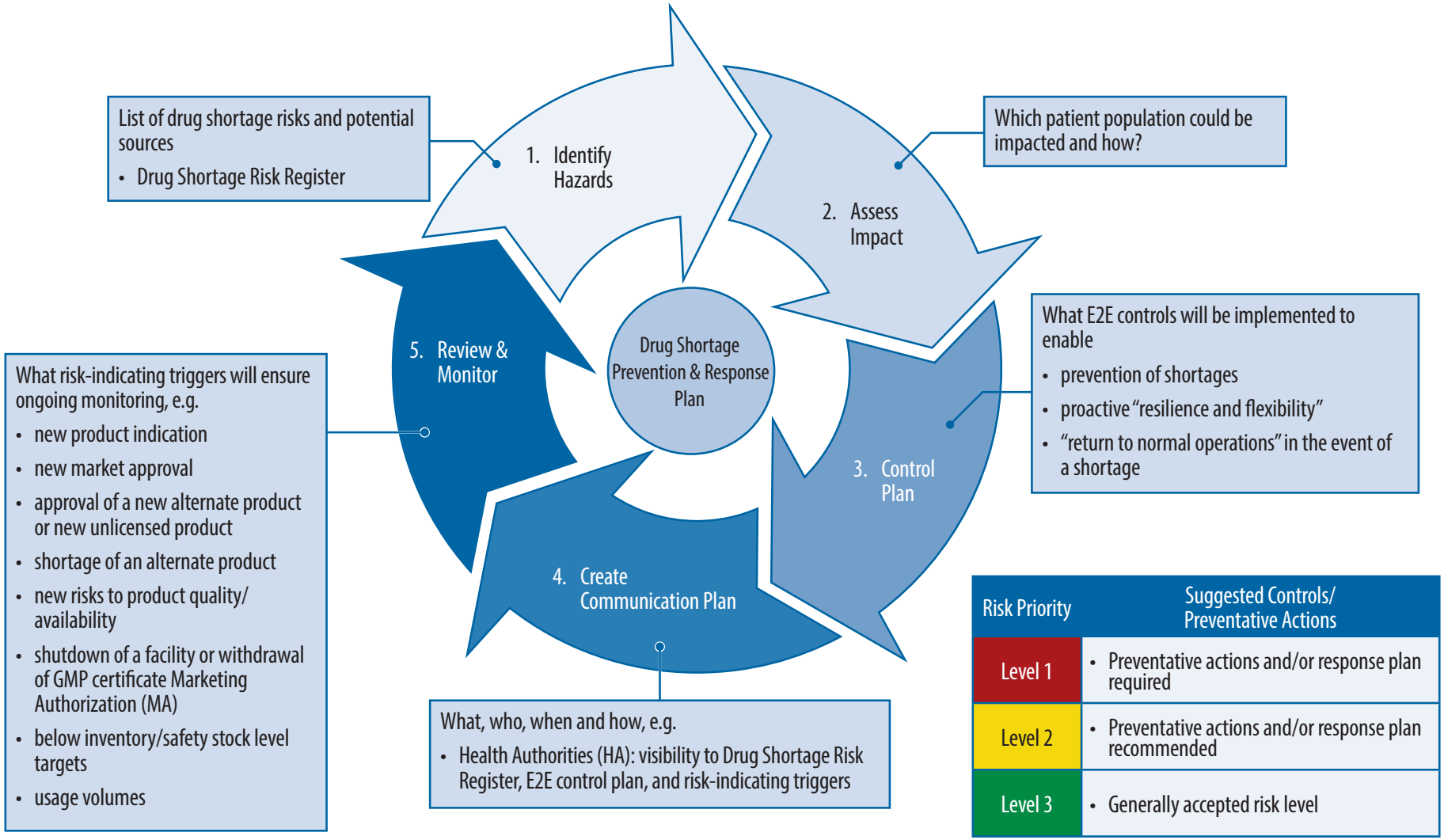

Risk Priority	Suggested Controls/ Preventative Actions
Level 1	• Preventative actions and/or response plan required
Level 2	• Preventative actions and/or response plan recommended
Level 3	• Generally accepted risk level

Figure 7.0-1 Suggested Controls and Preventive Actions for a Drug Shortage Prevention and Response Plan

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A Drug Shortage Prevention and Response Plan consists of the following information:

1. **Identification of drug shortage hazards:** An end-to-end risk assessment of the product's value chain until the first economic customer—its manufacturing operations, storage, and transportation, as shown in **Figure 7.0-2**—should identify potential sources of drug shortages. The Drug Shortage Prevention and Response Plan should address the management of the drug shortage risks listed in the Drug Shortage Risk Register. Typical visibility for a MAH ends after shipment to wholesale distributors.

2. **Assessment of the impact of a potential drug shortage:** Based on the product indications, an assessment should be made and documented of which patient population would be affected by a drug shortage and what this effect would be. Application of the risk triage model would identify the risk priority levels described in **Section 6.2**. In addition, clinical risk assessments would be helpful in understanding the clinical impact of a drug shortage.

3. **Risk control plan (short-term and long-term):** This section should describe the controls identified from the end-to-end risk assessments of the product, how the risk control strategy will proactively control drug shortage risks, what measures can be taken to ensure product supply resilience, and a plan to respond to a shortage and ensure a prompt return to normal operations in the event of a shortage.

4. **Communication plan:** The Drug Shortage Prevention and Response Plan should include a communication plan for drug shortages. The AESGP, EFPIA, EGA, and PPTA collaborative document provides guidance on communications related to product shortages, including possible triggers for communication strategies ***(38)***.

5. **Risk review and monitoring:** The Drug Shortage Prevention and Response Plan should be a living document that is reviewed and updated whenever a product's risk profile changes. Potential causes of a risk profile change for a product include approval of a new product or dosage form with the same or a similar indication, a new indication for the product, or a new market for the product; shutdown of a facility or withdrawal of its GMP certificate; increase in product usage volume or a drop in product inventory or safety stock to below target levels; and new quality or manufacturing issues. Proactive management of the Drug Shortage Prevention and Response Plan will ensure that shortages are mitigated in advance as much as possible.

Section 12.0: Appendix 2 provides a template for a Drug Shortage Prevention and Response Plan.

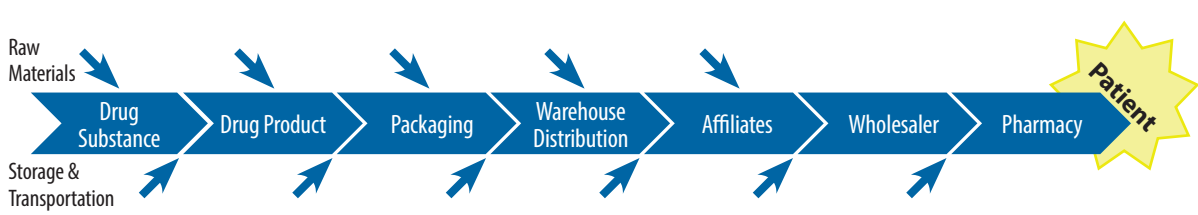

Figure 7.0-2 End-to-End Value Chain for a Product

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8.0 Stakeholder Management and Communications related to Drug Shortages

Multiple stakeholders are affected by, and need to be involved in, addressing a drug shortage—manufacturers, health authorities, distributors, pharmacies, healthcare providers and, sometimes, patients, depending on the cause(s) and level of the shortage. The legal obligations for MAHs to notify the relevant health authorities of a potential drug shortage are described in Article 13 of *Commission Directive 2003/94/EC* for the European Union and in Section 506C of the *Federal Food, Drug, and Cosmetic Act* for the United States. In addition, Part 1, Section 5.71 of the *EU GMP Guidelines* has been updated to provide guidance on notification of restrictions in supply. A collaborative dialog between the company and the relevant health authorities is key to facilitating the control and/or prevention of a shortage and, ultimately, ensuring a continued supply of medicinal products for patients.

Companies need to consider many things when reacting to a current or impending shortage of one of their medicines in a given country or region. Developing a harmonized risk communication mechanism to alert health authorities of a potential or existing shortage is important to creating a shared understanding of the shortage. The guidance paper *Quality and Manufacturing Driven Supply Disruptions: Industry Communication Principles to Authorities* provides communication principles and a framework to harmonize reporting and communications between MAHs and health authorities regarding drug shortages arising from quality and/or manufacturing disruptions. The paper proposes a harmonized trigger point, time point, and recipient(s) for communicating potential drug shortage events for nationally and centrally approved products in Europe ***(38)***.

Relevant health authorities, those where the product is marketed and where the shortage is expected to occur, must be notified when a meaningful disruption could occur in the supply of medicines due to manufacturing and quality issues, that is, when normal volumes of medicines cannot be delivered for further distribution. The risk matrix in **Table 6.1.1-1** categorizes risks to patients in a tiered mechanism for determining communication for the various risk levels based on the product's therapeutic use and the availability of alternative treatments. The level of effort and formality in the risk assessment for determining patient impact, resulting corrective actions, and level of communication with health authorities should be commensurate with the level of risk to patients.

A safety or clinical risk assessment should inform the risk communication strategy so that appropriate risk mitigation actions can be taken at the healthcare provider and/or patient level. For defective products (e.g., those with a potential lack of sterility assurance), a comparison might be needed of the clinical benefits and risks for patients of continued product use, especially when no alternative product or treatment is available. Partnership and expedited dialogue with health authorities in making appropriate risk mitigation decisions and returning to normal operations in such cases are important.

8.1 Key Risk-Indicating Information to Compile for Health Authorities

When anticipating a shortage may occur, companies should compile key pieces of information for health authorities when their involvement is needed. Knowing the reasons for the stock shortage or depletion not only facilitates the implementation of corrective and preventive actions (CAPA), but it can also help develop practical marketplace solutions for the shortage. Based on product ownership, it is important to openly clarify communication accountabilities between the health authority and the (1) manufacturer or first distributor; (2) wholesaler, agent, trader, distributor, repackager, or relabeler; and (3) suppliers of the product to the patient (e.g., retail pharmacies, hospital pharmacies, physicians).

The notification content and format described in the AESGP, EFPIA, EGA, and PPTA collaborative paper describe what data should be provided to health authorities for effective communications in the event of potential drug shortages. This includes details on the MAH, the product, the causes and details of the potential drug shortage, the risk level (A, B, or C) determined in the first step in the risk

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triage (**Figure 6.1-1**), the expected duration of the shortage, the steps taken or planned to address the shortage, the countries impacted, and the notification to other parties. Additional secondary information that can be helpful for the MAH to collect in order to reach a shared understanding of a shortage with the relevant health authorities and how best to address includes:

- Patient-Related Information
  - Types of patients (e.g., by disease, demographics, age, genetics, origin) using the medicine. A good level of clinical understanding is important to assess whether any patient groups may be particularly affected by the shortage (e.g., neonates, patients with cancer or diabetes).
  - Scale of the problem, for example, the number of patients likely to be affected, the duration of the shortage, and where the problem may be most acute. The following information can be useful to compile and assess:
    - Likely number of patients treated with the product each month and the percentage of patients that may be affected by the shortage or stock depletion. The current market share, usage profile of the product, and monthly volume of prescriptions for the product may also be useful to review.
    - Monthly sales volume of the product and the proportion of this volume that may be affected by the shortage or stock depletion
    - Possible local, regional, or global impact on product supplies

- Distribution Information in the Country or Region
  - Name and address of the primary distributor of the product, if any
  - If other distributors (i.e., secondary wholesalers) supply the product
  - If the product is supplied only to hospitals or to both hospital and retail pharmacies
  - If the product is supplied through other channels, such as clinics and nursing homes
  - Opportunities for parallel import, repackaging, and relabeling, depending on regulations

- Current Stock in Each Country or Region
  - Whether the product is currently out of stock in the country or region, or if another version of the product or process is available during the stock-out period
  - Whether the product stock is expected to be depleted in the near future (e.g., within the next few weeks or months) because of the supply issue
  - Whether the product is in stock but available quantities are insufficient to meet patient needs
  - Whether stockpiling is requested by the government, legislation, or guidance or as result of the Drug Shortage Prevention and Response Plan

This information can help determine the likely impact of the shortage on patients, if any, as well as the level of risk presented by the issue and whether the health authorities should be informed.

8.2 Risk-Based Decision-Making by the Company and Health Authorities

In the event of a shortage, having a product risk assessment developed proactively using the triage method and the risk matrices in **Section 6.1**, and the key risk-indicating information compiled in **Section 8.1** becomes very useful in determining the appropriate actions to take. In addition to the manufacturer or MAHs, CAPAs to address the risks may need to be owned by other stakeholders in the supply chain. Therefore, communication between the stakeholders, including, for example, affiliates of the MAH in a country or region, distribution centers, or pharmacies as well as the press, is most important to detect unintended consequences of a corrective or preventive measure early in the system.

Several decisions may need to be made by health authorities in response to reports of actual or pending product shortages:

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- Whether to become formally involved in managing the shortage or in coordinating a response to the shortage. The risk triage method described in **Section 6.1** will be useful in determining:
  - Whether health authorities need to become directly involved (e.g., by taking direct measures, coordinating responses, developing work plans and action items)
  - Whether becoming directly involved is not necessary for health authorities at that time
  - Whether health authorities need to monitor the situation closely without becoming directly involved

- Whether to facilitate the sourcing of alternative supplies of the product, for which the decision usually involves determining:
  - Whether the MAH should be allowed to import and supply a version of the product intended for a different market that may not be authorized in the country.
  - Whether this product could be batch packaged for the other market and supplied in that country's packaging and delivery. The packaging and product information may be in a different language than that of the country.
  - Whether the imported product needs to be repackaged and, if so, the level of regulatory approval and oversight required for the repackaging.
  - Whether the healthcare professionals should be advised that an unlicensed version of the product is available or that an unlicensed version of a different but clinically equivalent product is available.

- Whether to expedite approval of a request to use an alternate unlicensed facility or a post-approval change that has been submitted by the MAH. A related decision is whether the approval should be conditional, temporary, or permanent.

- What communication strategies will be required by the shortage or stock depletion, and how those strategies will ensure that companies, health authorities, healthcare professionals, and/or patients can take appropriate risk mitigation actions. After the clinical risk assessment is completed, the health authority may decide to advise healthcare professionals to take specific steps, such as:
  - Giving priority to certain groups of patients (e.g., patients already treated with the medicine) over others for continued treatment with the medicine
  - Using a different medicine or treatment for new (or all) patients
  - If a product is defective (e.g., loss of sterility assurance), weighing the clinical benefits and risks for patients of continued use of the product when no alternative product or treatment is available

Any time a shortage situation is encountered it should be viewed as an opportunity for the organization to review their risk register and, where relevant, assess suitable updates to their risk control program to improve prevention practices. This might involve examining the real root cause of why it occurred, modifying CAPAs to prevent recurrence of a shortage for that and similar products, and mitigating any gaps that the shortage event might have uncovered.

9.0 Conclusions

The objective of ensuring an uninterrupted supply of safe, efficacious products to patients is shared by regulators, legislators, healthcare providers, and industry bringing them together to communicate for jointly developing solutions that serve the best interests of patients. As the risk to the patient population from a drug shortage rises, the level of rigor, effort, and cross-functional collaboration within an

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organization to address the risk should also increase. This effort should be coupled with effective communication among the organization, health authorities, and other relevant stakeholders to proactively manage drug shortage risks.

Although challenging, the industry should address the concerns associated with drug shortages, given the objective of protecting patients from harm. Proactively identifying relevant drug shortage hazards and robustly managing drug shortage risks in the end-to-end product value chain, as well as effectively and promptly communicating potential shortages to all stakeholders, is essential to reduce the risk of drug shortages.

TR-68 provides a practical framework that companies can use, a triage model to assess drug shortage risks, and templates for developing a Drug Shortage Risk Register and a Drug Shortage Prevention and Response Plan at a product level. It also describes ways that industry can engage more proactively with health authorities to request expedited review and approval of certain PACs by multiple health authorities to reduce the risks of drug shortages. This could be achieved by using agreed-upon PAC management plans and harmonized gCPs. Such protocols would be helpful for companies to address the risk of drug shortages due to aging facilities, processes, and analytical technologies by modernizing their facilities and implementing new technologies. This technical report also leverages the guidance developed by the AESPG, EFPIA, EGA, and PPTA collaboration for communications between MAHs and health authorities.

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11.0 Appendix 1

Company:	**Drug Shortage Risk Register** *(Template as part of TR-68 (Revised), ©2024 PDA, Inc.)*	***Date:***

ID	Product and Strength	Potential Drug Shortage Hazard	Hazardous Situations	Countries/ Regions that Could be Affected	Affected Internal Area	Initial Risk Level	Risk Indicators	Planned Risk Control Actions	Risk Level After Risk Control	Person(s) Responsible	Status	Due Date
Unique reference number	*Identify product and strength*	*Risk description in a newspaper headline style; reference as applicable*	*Root cause or risk assessment title*	*Identify countries/ regions that could be affected*	*All sites/locations, affiliates, and functional areas affected by a shortage*	*Risk Level or indication of the risk as high or medium. Note: Based on a consideration of current shortage prevention controls and their likely effectiveness*	*Potential leading and lagging indicators for this drug shortage risk*	*Planned mitigation actions, (CAPA number, if applicable)*	*Risk Level after mitigation or whether residual risk is high or medium*	*Who is responsible for coordination of the mitigation action(s)*	*"Planned," "In process," "Delayed," or "Completed"*	*Current deadline for completing recommended actions [dd.mm.yy]*

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12.0 Appendix 2

Drug Shortage Prevention and Response Plan

Risk Triage Approach for Proactive Management of Drug Shortage

(Template is part of TR-68 (Revised), © 2024 PDA, Inc.)

Complete this triage model for each dosage form by assessing the product categorization, the impact of the shortage on patients, and the end-to-end controls used to prevent or mitigate the risk. Use the form proactively to identify ways to prevent a shortage or reactively to respond to a shortage.

A. Basic Data

API name		Drug Product Name	
Pharmaceutical Form and Strengths		ATC (Anatomical Therapeutic Chemical) Code	
Pack Size(s)		Therapeutic Indication(s)	
Route(s) of Administration		Contact Person/Contact Information	
Other Information[1]			

[1] Other information available may include, for example, countries where marketed, reference to market authorization; in case of response plan: affected by the supply disruption, shortage details including impacted countries, expected end date.

B. Impact to Patient

Mark the Risk Level as **A**, **B**, or **C** based on the product's therapeutic use and availability of alternative therapies using the criteria below.

			Availability of Alternatives		
			No Alternatives Available	Alternative Products Available/API Available	Same Product/API Available
Therapeutic Use & Consequences if Alternate Product not Available	Medically necessary product, life supporting or life sustaining	Fatal or severe irreversible harm if the patient is not treated with the product	Risk Level A	Risk Level A	Risk Level B
	Acute short term or chronic long term	Severe harm but reversible if patient is not treated with the product	Risk Level A	Risk Level B	Risk Level C
	Other indications	Inconvenience if patient is not treated with the product	Risk Level B	Risk Level C	Risk Level C

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C. Risk Priority Level (1, 2, or 3) Based on Likelihood of a Shortage

Based on the Risk Level (A, B, or C) identified in **Section B,** the probability of a shortage, and the consideration of the drug shortage prevention controls currently in place and their likely effectiveness, indicate the Risk Priority Level (1, 2, or 3) using the criteria below.

		Likelihood of Shortage		
		High	Medium	Low
Therapeutic Use & Consequences if Product not Available	Risk Level A	Risk Priority Level 1	Risk Priority Level 1	Risk Priority Level 2
	Risk Level B	Risk Priority Level 1	Risk Priority Level 2	Risk Priority Level 3
	Risk Level C	Risk Priority Level 2	Risk Priority Level 3	Risk Priority Level 3

D. Risk Control Activities for a Proactive Drug Shortage Prevention Plan

Based on the potential Risk Priority Level (1, 2, or 3), describe suggested controls to be established, the communication plan (in the event of a shortage), and other considerations. This can be done either in text or table form. See a potential table format below.

Risk Priority	Suggested Controls / Preventive Actions
Level 1	…
Level 2	…
Level 3	Generally accepted risk level

E. Risk control Activities for a Drug shortage Response Plan

Develop a drug shortage response plan for implementation in the event of a drug shortage and discuss the plan, as needed, with the relevant health authorities.

1. Identify the reason(s) for the shortage or temporary disruptions (check all that apply):
   - Manufacturing Issues
     - ☐ Industry, e.g., capability, capacity, storage, transport/distribution/shipment, regulatory and quality compliance, CDMO
     - ☐ Component, e.g., raw materials, API, excipient, equipment
     - ☐ Government, e.g., changes in legislation or guidance, enforcement, intellectual property rights protection
     - ☐ Multistakeholder, e.g., infrastructure (e.g., power supply), registration process
     - ☐ Time to manufacture and batch release

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- Business Issues
  - ☐ Unexpected product demand
  - ☐ Industry, e.g., quotas
  - ☐ Regulatory delay
  - ☐ Wholesale, distribution, e.g., parallel trade, demand, known real manufacturer
  - ☐ Government, e.g., national protectionism, export ban, health economics, channel strategies
  - ☐ Multistakeholder, e.g., commercial priorities, stockpiling, number of competitors
- Economic and Geographic Issues
  - ☐ Industry, e.g., market withdrawals, delayed introduction
  - ☐ Wholesale, distribution, e.g., commercial tendering
  - ☐ Governments, e.g., price reductions, tax structure, reference pricing, reductions in spending
  - ☐ Multistakeholder, e.g., exchange rate, tightening of payment, payment delay, affordability
- Reliability Issues
  - ☐ Wholesale, distribution, healthcare provider, e.g., off-label use, compassionate use, competitor sourcing
  - ☐ Governments, e.g., environmental standards
  - ☐ Multistakeholder, e.g., competent staff, illegal product introduction, theft, stolen material, counterfeits, substandard, national disasters, health emergencies, pandemic, demand increase
- Others, please specify _____

2. Assess risks to patients from the shortage per **Section B** (attach risk assessment), including the expected duration of the situation and the patient groups likely to be affected by the shortage (e.g., neonates, people with diabetes).

3. Create (and attach) a short-term action plan, based on **Section D** and/or discussions with relevant health authorities, to include actions that can be taken to address the shortage in the near term (e.g., when alternative therapies are available).

4. Execute the communication plan based on the shortage's characteristics (e.g., Dear Health Care Provider letters, informed consent forms, or patient letters)

5. Create (and attach) a longer-term action plan based on **Section D** above, including actions and timelines needed to return the product supply to normal.

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F. Risk Review and Updates

Identify the triggers for reviewing and updating the Drug Shortage Prevention and Response Plan.

Check the trigger(s) that will drive reviews and revisions of the Drug Shortage Prevention and Response Plan:

- ☐ New product indication
- ☐ New market approval
- ☐ Approval of a new alternative or unlicensed product
- ☐ Shortage of an alternate product
- ☐ New risks to product quality or availability
- ☐ Shutdown of a facility or withdrawal of GMP Certificate
- ☐ Inventory or safety stock levels below target level
- ☐ Increase in product usage volumes
- ☐ Others, please specify _____

G. Approvals Required

Name	Title and Function	Signature	Date
	Person Responsible		
	QA Decision-Maker		
	Senior Management Function		

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Risk-Based Approach for Prevention and Management of Drug Shortages

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